

The Paradox of Opportunity: Socioeconomic Origins of Financial Gender Gaps*

Athena Tsouderou[†] Şelale Tüzel[‡]

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Abstract

We study gender gaps across two central domains of household finance: stock market participation and mortgage borrowing and refinancing. Using rich microdata, we show that these gaps are small among lower-income, less-educated households but widen sharply at higher socioeconomic status, with women participating less in the stock market, borrowing on less favorable terms, and refinancing less optimally than comparable men. Financial literacy grounded in quantitative reasoning explains much of this gradient. Leveraging within-family sibling comparisons, we trace the gaps to childhood differences in math skills and math-related attitudes shaped by persistent gender stereotypes.

Keywords: Financial literacy, stock market participation, mortgages, math skills, gender gap.

JEL Classification: D14, G11, G21, G53, J16.

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[†]Herbert Business School, University of Miami, email: atsouderou@miami.edu.

[‡]Marshall School of Business, University of Southern California, email: tuzel@usc.edu.

1 Introduction

Household balance sheets are shaped by a few consequential financial choices. Two of the most important are whether to participate in equity markets and how to finance housing through mortgage borrowing. These decisions determine exposure to the equity risk premium, the cost of housing, and the returns on the largest assets held by most households. As a result, they play a central role in shaping long-run wealth accumulation and economic welfare.

A large literature documents persistent gender differences in financial behavior and outcomes. Women are less likely than men to hold stocks, tend to take less financial risk, and often experience less favorable outcomes in credit markets (Jianakoplos and Bernasek, 1998; Sunden and Surette, 1998; Barber and Odean, 2001; Lusardi and Mitchell, 2014). Common explanations point to differences in preferences, beliefs, financial sophistication, or access to financial markets. A prevailing intuition, consistent with these views, is that gender gaps should narrow as opportunities expand and financial knowledge deepens, particularly among higher-income and more educated households, where access to markets and information is broad. This logic is especially salient in an era of rapid financial innovation: online trading platforms, robo-advisors, and fintech tools have sharply reduced information and participation costs, which, in principle, should contribute to more equal financial outcomes.

This paper shows that the opposite pattern holds. We document a paradox of opportunity: while financial outcomes improve with socioeconomic status for both men and women, the gains are larger for men. As a result, gender gaps in two central household finance outcomes, stock market participation and mortgage borrowing, are small among lower-socioeconomic-status (SES) households but widen substantially among higher-income and more educated households. This pattern runs counter to the view that greater resources, information, and market access should narrow gender gaps. It suggests that modest average gender differences conceal substantial heterogeneity across socioeconomic environments, an aspect that has received limited attention in prior studies of gender gaps in investment and credit markets.

A central feature of our analysis is that we study this pattern across financial decisions that differ sharply in institutional setting but share a common element: households must recognize, evaluate, and act on financially valuable opportunities. The recurrence of the same gender-by-SES pattern in stock market participation and mortgage decisions suggests that the widening gaps are unlikely to reflect a market-specific friction and instead point to a broader mechanism shaping household financial decision-making.

We begin by studying stock market participation using data from the Panel Study of Income

Dynamics (PSID). While gender gaps are small among individuals from lower socioeconomic backgrounds, they widen sharply with socioeconomic status: women participate substantially less than men at higher levels of income and education. We show that this pattern is not explained by gender differences in risk tolerance, wealth, or expected future economic circumstances, including anticipated income and household composition. Instead, financial literacy emerges as a key factor. We establish a new empirical fact: gender gaps in financial literacy widen with socioeconomic status, and this widening accounts for a substantial share of the increase in stock market participation gaps among higher-income and more educated households.

The importance of financial literacy raises the question of where these literacy differences originate. Financial decisions such as stock market participation and mortgage choice require individuals to reason about compounding, risk–return tradeoffs, discounting, and intertemporal comparisons. As a result, literacy in these domains depends disproportionately on quantitative reasoning skills rather than on general cognitive ability alone.

We therefore turn to math-related skills and attitudes earlier in life. Prior research in education and social norms shows that gender differences in quantitative domains can be more pronounced in socioeconomically advantaged environments, making childhood math achievement and math-related attitudes a natural place to look (Breda et al., 2020; Dossi et al., 2021). Using the Child Development Supplement (CDS) of the PSID, we examine whether these childhood differences help explain adult financial behavior. The CDS allows us to follow children into adulthood and to compare brothers and sisters raised in the same family. This design holds constant family background and parental resources and enables us to distinguish math-specific skills from general cognitive ability by controlling for reading achievement, a common proxy for overall cognition.

Using the CDS and a within-family sibling comparison design, we show that boys are more likely than their sisters to participate in the stock market as adults, with a gender gap of 3.2 percentage points in low-SES families that increases by an additional 10.2 percentage points in high-SES families. Differences in childhood math achievement account for about 20 percent of the additional participation gap within high-SES families, whereas reading achievement has no marginal explanatory power. Moreover, gender differences in math attitudes, measured by children’s confidence in and enjoyment of math, widen with socioeconomic status even after controlling for achievement, consistent with stronger gender-biased math norms in higher-SES families, and these attitudes predict subsequent math performance. Together, these results trace a pathway from math-related attitudes to math skills and, ultimately, to adult financial behavior, with divergence between boys and girls emerging most strongly in socioeconomically advantaged families.

We next examine whether these patterns extend to high-stakes mortgage decisions later in life. Mortgage markets provide a natural setting for studying socioeconomic heterogeneity because borrowers across the income distribution play distinct and economically important roles in mortgage originations and defaults (Adelino et al., 2016). While the PSID is well suited for studying stock market participation by linking siblings’ childhood experiences to early adult financial behavior, mortgage outcomes occur later in life and require much larger samples to account for timing, location, and detailed borrower, loan, and property characteristics. We therefore use a near-universe sample of U.S. conventional mortgages constructed by merging Home Mortgage Disclosure Act (HMDA) data with loan-level information from Fannie Mae and Freddie Mac.

We show that single female borrowers obtain less favorable mortgage terms and refinance less frequently when rates decline, with gaps that are largest in higher-income and more educated areas. These differences have economically meaningful consequences: under conservative assumptions, we estimate that gender gaps in mortgage pricing and refinancing raise 10-year borrowing costs for single women by roughly \$3,000 in higher-SES areas, compared with about \$1,200 in lower-SES areas. These costs arise from a combination of less favorable initial mortgage terms and lower refinancing rates. Consistent with our stock market findings, regions with larger male-favoring math gaps among children and adults also exhibit larger gender gaps in mortgage outcomes, reinforcing a common mechanism across financial domains.

A natural concern in interpreting the socioeconomic gradient is that smaller gender gaps in lower-SES settings may reflect compressed choice sets or limited scope for financial decisions to vary, rather than a true widening of gender differences with socioeconomic status. We address this concern in both domains. In the stock-market analysis, the results are robust to excluding households at the bottom of the income and education distributions, where participation may be most constrained. In the mortgage analysis, we examine whether mortgage terms are less dispersed in low-SES tracts. They are not. For several key pricing margins, mortgage terms are in fact more dispersed in low-SES areas. Thus, the larger gender gaps in high-SES areas do not appear to be a mechanical consequence of greater dispersion in those markets.

Both stock market participation and mortgage choice involve complex financial tradeoffs, and prior work shows that mistakes in these domains are closely linked to numeracy and financial sophistication (Gerardi et al., 2013; Agarwal et al., 2016; Keys et al., 2016). The mid-2000s housing boom saw widespread use of complex mortgage contracts with adjustable rates and non-standard repayment structures (Amromin et al., 2018), increasing the importance of financial sophistication in evaluating long-term borrowing costs. Differences in financial sophistication are not merely individual frictions but can generate meaningful redistribution: heterogeneous

refinancing behavior creates cross-subsidies in mortgage markets, effectively transferring surplus from less sophisticated to more sophisticated borrowers (Fisher et al., 2024; Berger et al., 2024). Against this backdrop, we provide complementary evidence using data from the National Survey of Mortgage Originations (NSMO) that gender differences are particularly pronounced for advanced financial literacy, such as understanding rate–point tradeoffs and amortization dynamics, rather than for basic financial concepts. While advanced financial literacy rises sharply with income and education for men, this gradient is substantially flatter for women, contributing to widening gender gaps in high-SES environments, a pattern we consistently observe across datasets.

Our findings contribute to several literatures. First, we contribute to the literature on stock market participation and household portfolio choice, which emphasizes participation costs, background risk, beliefs, life-cycle income, household-composition dynamics, and financial literacy as determinants of market entry (Haliassos and Bertaut, 1995; Vissing-Jørgensen, 2002; Guiso et al., 2002; Gomes and Michaelides, 2005; van Rooij et al., 2011; Bacher, 2025; Bucher-Koenen et al., 2025). Gender differences in participation and risk-taking have been widely documented (Jianakoplos and Bernasek, 1998; Barber and Odean, 2001). We extend this work by showing that gender gaps in participation are not uniform but increase sharply with socioeconomic status.

Second, we contribute to research linking cognitive and quantitative skills to financial behavior. Prior work shows that cognitive ability is strongly associated with stock market participation and portfolio performance (Grinblatt et al., 2011), and that quantitative ability, in particular, is closely tied to the quality of financial decision-making, whereas verbal ability adds little explanatory power (Agarwal and Mazumder, 2013). A related literature emphasizes that quantitative skills are distinct from general cognitive ability (Gerardi et al., 2013), are central for financial and labor market outcomes (Goodman, 2019; Brown et al., 2016; Cole et al., 2016), and that financial education programs that do not explicitly target numeracy have limited effects on financial decision-making (Carpena et al., 2011). Consistent with this evidence, Lusardi (2012) documents widespread deficits in numeracy, particularly among women, while Adams et al. (2023) show that the math gender gap predicts women’s representation in investment professions across countries and U.S. states. Complementing this work, Altmejd et al. (2024) provide causal evidence that enrollment in business and economics programs improves financial outcomes, consistent with financial education and math skills being mutually reinforcing. We build on this literature by showing that math-specific skills and attitudes formed early in life play a distinct role in explaining gender gaps in financial outcomes and, importantly, why these gaps widen with socioeconomic status.

Third, we contribute to the literature on mortgage choice and refinancing, which documents that many households pay unnecessarily high borrowing costs and fail to refinance optimally when interest rates fall (Agarwal et al., 2016; Keys et al., 2016; Bhutta et al., 2026; DeFusco and Mondragon, 2020). We show that systematic gender differences persist even in the modern U.S. mortgage market and that these differences widen with socioeconomic status.

Related work documents gender differences in housing investment returns. Goldsmith-Pinkham and Shue (2023) show that single women earn lower returns on housing investments than men, driven by differences in market timing and negotiation outcomes, while Andersen et al. (2021) find no gender differences in negotiation outcomes once housing preferences are accounted for. Our results complement this literature by showing that gender gaps in housing returns can also arise through the financing side of housing transactions, via differences in mortgage pricing and refinancing behavior.

Our results are also related to work on discrimination and differential treatment in financial settings. This literature shows that gender gaps arise in mortgage appraisals, career advancement and disciplinary outcomes within financial institutions, and financial advice (Bosshardt et al., 2026; Huang et al., 2024; Egan et al., 2022; Bucher-Koenen et al., 2025). Our setting points to a complementary mechanism: gender gaps are largest in high-SES environments and in decisions that require households to recognize, evaluate, and act on financial opportunities, highlighting the role of financial literacy and quantitative reasoning.

The mortgage lending literature also documents substantial racial disparities in borrowing outcomes. Prior work shows that racial minorities often face higher interest rates and borrowing costs, differences frequently linked to credit risk, market segmentation, or lending constraints (Ghent et al., 2014; Bhutta and Hizmo, 2021; Bartlett et al., 2022). In contrast, the gender gaps we document persist despite lower default rates among women and are largest in more socioeconomically advantaged areas, pointing to a distinct set of underlying mechanisms.

Finally, our paper connects household finance to the education literature on gender norms and quantitative domains. Prior work documents larger gender gaps in math achievement and math-related attitudes in socioeconomically advantaged settings, particularly among affluent white families, and attributes these patterns to social norms and gender stereotypes (Reardon et al., 2019; Dossi et al., 2021; Breda et al., 2020). We build on this literature by showing that these patterns matter beyond schooling and occupational choice: they help explain gender gaps in consequential household financial decisions and why those gaps widen with socioeconomic status.

The remainder of the paper proceeds as follows. Section 2 describes the PSID data and

presents results on gender gaps in stock market participation, including their connection to financial literacy, childhood math skills and attitudes, and related evidence on gender norms. Section 3 describes the mortgage data, presents evidence on gender gaps in mortgage pricing and refinancing behavior, and examines potential mechanisms underlying these gaps, including links to financial literacy and math skills. Section 4 concludes.

2 Stock Market Participation

2.1 PSID Data

We use data from the Panel Study of Income Dynamics (PSID), a nationally representative longitudinal survey of U.S. households. Our analysis draws on the PSID Main Family Files, the Child Development Supplement (CDS) and the Transition into Adulthood Supplement (TAS). The CDS provides rich measures of children’s cognitive skills and attitudes, including standardized math test scores and math-related attitudes, collected in 1997, 2002, and 2007. We link these childhood measures to adult financial outcomes observed in the PSID Main Family Files (1996–2023) and TAS (2005–2021), which contain detailed information on education, income, and asset ownership, including stock market participation. We also incorporate a measure of financial literacy from the 2016 Wellbeing and Daily Life Supplement and a measure of risk tolerance derived from the Risk Tolerance Project questions administered in 1996. The longitudinal structure of the PSID allows us to connect childhood math skills and attitudes to financial decisions in adulthood while accounting for shared family background through sibling comparisons.

2.2 Summary Statistics

Table 1 summarizes the key variables used in the analysis. The top panel reports descriptive statistics for the sample of single adult household heads drawn from the PSID Family Files. The bottom panel reports statistics for the sample constructed by linking the CDS, TAS, and Family Files, which follows individuals from childhood into early adulthood as single adults.

We classify adults as having high socioeconomic status (SES) if they have completed at least one year of postsecondary education (more than 12 years of schooling) and have income above the median in the single-adult sample; all others are classified as low SES. Under this definition, 65 percent of adults are low SES and 35 percent are high SES. Women account for

53 percent of unique individuals but 58 percent of observed financial outcomes, reflecting longer life expectancy and a higher likelihood of being single at older ages; on average, women in the sample are approximately six years older than men. Educational attainment is similar for men and women, about 12 years in the low-SES group and 15 years in the high-SES group, while men earn roughly \$10,000 more on average. Gender differences in safe assets (total wealth, including housing equity, but excluding investments in stocks and IRAs) are modest among low-SES individuals, with men holding 7 percent more than women. In contrast, among high-SES individuals, men hold 43 percent more safe assets than women.

Our PSID measure of financial literacy is based on a single compound-interest question, one of the canonical “Big Three” items (Lusardi and Mitchell, 2014). We interpret this measure as capturing a numeracy-intensive component of financial literacy rather than the full breadth of financial knowledge. This distinction is important for our setting, because prior work shows that numeracy and interest-rate calculations are central components of financial literacy and are among the strongest predictors of financial behavior (Lusardi and Tufano, 2015; Song, 2020). These measures are also strongly correlated with broader literacy indices, suggesting that simple compound interest questions capture a substantial share of the variation in overall financial knowledge (van Rooij et al., 2011). We find that financial literacy differs sharply by gender, with gaps that widen at higher socioeconomic status. Among low-SES adults, 10 percent of women and 18 percent of men answer the question correctly, while among high-SES adults the corresponding shares are 30 percent for women and 50 percent for men. Men exhibit slightly higher average risk tolerance than women, and risk tolerance is modestly higher in the high-SES group overall. Stock market participation is 5 percent for both men and women in the low-SES group, and 17 percent for high-SES women and 22 percent for high-SES men.¹

Our child-to-adult panel begins with the Child Development Supplement (CDS), which tracks children in PSID families who were ages 0–12 in 1997 across three waves from 1997 to 2007. Each child is observed up to three times through age 18 from CDS, and subsequently followed into adulthood in later PSID waves. We restrict the sample to children living with a parent who is the household head and for whom education information is available for both parents.² We classify families as high SES if both parents have at least one year of postsecondary education and family income exceeds the sample average; all other families are classified as low SES. Under this definition, approximately one quarter of children come from high-SES families.

¹For reference, the Survey of Consumer Finances reports direct stock ownership rates of 15% in 2019 and 21% in 2022 for U.S. families (Aladangady et al., 2023). The SCF estimates are reported for all families rather than only single individuals.

²Missing parental education almost always reflects the absence of a parent or step-parent from the household. We focus on two-parent households to separate socioeconomic status from family structure.

The average child is just over 11 years old at the time of observation. Parents in high-SES families have an average of 15.5 years of schooling, about three years more than parents in low-SES families, and average family income is approximately \$105,000 in high-SES families compared with \$40,000 in low-SES families.

The CDS surveys children along multiple dimensions, including their abilities and achievement in math and reading, as well as their attitudes toward these subjects. Attitudes are captured through Eccles' self-concept measures of confidence, enjoyment, and perceived ability. A higher attitude score reflects stronger self-beliefs, greater intrinsic interest, and higher perceived competence in the relevant domain (math or reading). Achievement is measured using standardized math and reading tests. We convert math and reading test scores into within-sample deciles. Children from high-SES families score roughly two deciles higher than those from low-SES families in both math and reading. Gender differences in math achievement are modest among low-SES children, with boys scoring slightly higher than girls (0.16 deciles), but widen substantially in high-SES families, where boys score 0.52 deciles higher than girls. Consistent with this pattern, boys from high-SES families report significantly stronger math attitudes than all other groups, scoring approximately 0.3 standard deviations above the sample mean. In contrast, girls from high-SES families report slightly lower math attitudes than girls from low-SES families, despite ranking 2.2 deciles higher in math achievement. At the same time, they report strong attitudes toward reading, consistent with their higher reading achievement, indicating that the divergence in attitudes is specific to math rather than reflecting a broader confidence gap. This pattern foreshadows the mechanism we examine below: in high-SES families, girls' weaker identification with math appears despite strong academic performance, consistent with math-specific gender norms rather than lower overall ability or confidence.

We follow CDS children into adulthood through the Transition into Adulthood Supplement (TAS) and subsequently the PSID Family Files once they form their own households and remain single. Individuals in this sample are younger than the full adult sample, with an average age of about 24, and there is no age difference by gender. Stock market participation is higher among individuals from high-SES family backgrounds, with a substantially larger gender gap in this group: participation rates are 4 percent for women and 7 percent for men from low-SES backgrounds, compared with 17 percent for women and 23 percent for men from high-SES backgrounds. Despite differences in age and sample construction, these participation rates are similar to those observed in the full adult sample reported in the top panel of Table 1.

2.3 Gender Gaps in Stock Market Participation

Our empirical strategy examines gender gaps in stock market participation and how these gaps vary across the socioeconomic status, measured by income and education. We estimate linear probability models that regress individual stock ownership on an indicator for women interacted with alternative measures of socioeconomic status. To obtain our first set of results for the full adult PSID sample, we estimate the following specification:

$$\text{OwnStock}_{it} = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{SES}_{it} + \beta_3 \text{Female}_i \times \text{SES}_{it} + \gamma Z_{it} + \alpha_t + \epsilon_{it}, \quad (1)$$

where OwnStock_{it} is a dummy variable equal to 1 if individual i , who is the household head, owns “any shares of stock, mutual funds, or investment trusts not held in IRAs or other retirement accounts” in survey wave t , and 0 otherwise. Female_i is an indicator equal to 1 if individual i is a woman and 0 otherwise. SES_{it} denotes socioeconomic status, which we measure using three alternative proxies: log family income, years of education of the individual, and a SES indicator equal to 1 if the individual has completed more than 12 years of schooling and has above median income. All specifications include survey wave fixed effects, denoted by α_t , and control for age, income, and education, denoted by Z_{it} . Appendix Table A1 provides detailed definitions of the variables used in the PSID analysis.

Table 2 reports estimates from equation (1). Column (1) shows that women’s stock market participation is 3.4 percentage points lower than men’s after controlling for income, education, and age. Columns (2) and (3) interact the female indicator with standardized log income and education, respectively, and show that the participation gap increases with both variables: a one-standard deviation increase in income or education is associated with an additional participation gap of approximately 2 percentage points. Column (4) interacts the female indicator with the SES indicator and shows that the gender gap is 2.1 percentage points among low-SES individuals and rises to 6.2 percentage points among high-SES individuals.

To shed light on the mechanisms underlying the gender gap, we augment the baseline specification with measures of wealth, risk tolerance, and financial literacy. Because risk tolerance and financial literacy are observed in only one survey wave, 1996 and 2016, respectively, their inclusion substantially reduces the estimation sample. For individuals with non-missing values, we carry the observed measure forward to subsequent survey waves in which they are observed, but do not assign it to earlier waves. To facilitate meaningful comparisons across specifications, we re-estimate the baseline model, including the $\text{Female} \times \text{SES}$ interaction term, on each restricted subsample and report the corresponding results in columns (5), (7), and (9).

The descriptive statistics in Table 1 show that high-SES men hold substantially more safe assets than high-SES women. To assess whether this wealth gap accounts for the participation patterns, column (6) of Table 2 adds safe asset wealth to the baseline specification. Although safe asset wealth is a strong predictor of stock market participation, its inclusion has only a minor effect on the gender participation gap and its variation across SES groups, suggesting that wealth differences are not the primary driver of these patterns.

Column (8) adds the imputed risk tolerance measure and shows that, similar to wealth, risk tolerance is strongly related to stock market participation. However, despite the small average gender difference in risk tolerance documented in Table 1, controlling for risk tolerance explains virtually none of the gender gap in participation or its socioeconomic gradient: the coefficients on the female indicator and its interaction with SES are essentially unchanged.

A recent paper by Bacher (2025) proposes a life-cycle explanation for gender differences in stock market participation, showing that preference differences are not necessary to generate participation gaps. In her model, gender-specific trajectories of income and household composition can lead to different optimal portfolio choices even when men and women have identical preferences. This mechanism could potentially contribute to the SES gradient we document if expected future income or household composition vary systematically by gender and socioeconomic status. To assess this possibility, Appendix Table A2 estimates gender-specific forecasting equations for future income and household size. In line with Bacher (2025), future economic circumstances are highly predictable from current characteristics, and the relationships differ across genders. For example, future income loads positively on SES for both men and women, but the relationship is substantially stronger for women.

Using the fitted values from these forecasting equations, Appendix Table A3 adds controls for expected future income and household size. Consistent with Bacher (2025)’s mechanism, expected future income is strongly associated with stock ownership and substantially reduces the unconditional gender gap: in column (3), the female coefficient falls from -0.034 to -0.008 and becomes statistically insignificant. However, these controls do not explain the socioeconomic gradient in the gender gap. If anything, the gradient becomes more pronounced: the Female $\times$ SES coefficient in column (4) increases in magnitude from -0.030 to -0.058 after controlling for expected future income and household size. Thus, expected future economic circumstances help explain why women participate less in the stock market on average, but not the central pattern in our paper, the widening of the gender gap with socioeconomic status. We therefore next examine whether differences in financial literacy can help account for this SES gradient.

Column (10) of Table 2 adds the financial literacy indicator and shows that stock market

participation is strongly associated with literacy: financially literate individuals are 13.5 percentage points more likely to participate. Including financial literacy reduces the coefficient on the female indicator by nearly one half and the Female $\times$ SES interaction by roughly one third, indicating that gender differences in financial literacy account for a substantial share of the participation gap. This finding naturally raises the question of where these disparities originate. The PSID financial literacy measure, the compound interest question, captures a concept central to many financial decisions and relies directly on mathematical understanding, suggesting that these literacy gaps may reflect deeper differences in quantitative skills and attitudes formed earlier in life. Motivated by this connection, we next turn to the Child Development Supplement to examine whether gender differences in childhood math attitudes and achievement help explain the adult participation patterns we document.

Before turning to childhood origins of these literacy differences, we address two related concerns about the interpretation of the SES gradient in stock market participation. The first is that the small gender gap among low-SES households may reflect limited financial capacity rather than similar behavior: households at the bottom of the income distribution may not participate in the stock market regardless of gender. To address this, we construct a subsample excluding households below the 25th percentile of the income and education distributions. This restriction increases average income by more than \$10,000, raises participation among low-SES individuals, and yields a more balanced SES distribution (Appendix Table A4). Appendix Table A5 shows that the Female $\times$ SES interaction is virtually unchanged in this subsample. The results are also robust to excluding the bottom third of the income distribution, indicating that the widening gap at higher SES is not driven by mechanical nonparticipation among the poorest households.

A related concern is that the widening gap could reflect scaling properties of the linear probability model, given low baseline participation at the bottom of the distribution. We therefore re-estimate the main specification using nonlinear probability models. In the logit specification (Appendix Table A6), the Female $\times$ SES interaction remains negative and statistically significant. In terms of average marginal effects, women are 1.4 percentage points less likely than men to participate in low-SES households and 3.3 percentage points less likely in high-SES households. The interaction coefficient of -0.23 corresponds to an odds ratio of approximately 0.8, implying that the relative odds of participation for women in high-SES households are about 20 percent lower than predicted by the low-SES gender gap alone. A probit model yields nearly identical marginal effects. Together, these results confirm that the increasing gap reflects a genuine divergence in participation behavior rather than a floor effect or scaling artifact.

2.4 Childhood Math Skills, Attitudes, and Future Stock Market Participation

To examine how childhood math skills and attitudes shape stock market participation later in life, we focus on individuals in the PSID who are observed both in childhood and in young adulthood and adulthood. We estimate a series of specifications linking gender and socioeconomic status (SES) to adult stock market participation, and relate these outcomes to measures of math achievement and math-related attitudes formed in childhood. Our identification strategy exploits within-family variation by comparing sisters and brothers raised in the same household, allowing us to isolate gender differences from unobserved family background characteristics.

We begin by confirming our earlier findings on gender gaps in stock market participation (column (4) of Table 2) in the child-to-adult panel. Specifically, we estimate the following specification with family fixed effects:

$$\text{OwnStock}_{it} = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Female}_i \times \text{SES}_{i,1997} + \gamma Z_{it} + \alpha_t + \delta_f + \epsilon_{it}, \quad (2)$$

where OwnStock_{it} is an indicator for stock ownership in adulthood in year t , and Female_i indicates gender for individual i . $\text{SES}_{i,1997}$ equals one if both parents completed at least one year of postsecondary education and family income exceeded the median in 1997, the first year of the CDS. The specification controls for age in year t (Z_{it}), survey wave fixed effects (α_t), and family fixed effects (δ_f). Identification comes from differences between siblings within the same family, and standard errors are clustered at the family level.

Table 3 reports estimates from equation (2). Column (1) shows that gender gaps in stock ownership are small in low-SES families (3.2%) but substantially larger in high-SES families (13.4%), based on comparisons between siblings raised in the same household. In column (2) we augment the specification with indicators for being in the top decile of childhood math and reading achievement. Being in the top math decile is strongly associated with higher stock market participation, whereas being in the top reading decile is not, suggesting that particularly strong quantitative reasoning skills, rather than general cognitive ability, are most relevant for financial participation. The coefficient on the $\text{Female} \times \text{SES}$ interaction declines relative to column (1), indicating that differences in childhood math achievement account for a non-trivial share of the larger gender gap in stock market participation observed among high-SES families.

Column (3) examines gender differences in childhood math achievement directly. Among low-SES families, we find no meaningful gender difference in math test performance. In contrast, a sizable gap emerges in high-SES families: girls score nearly one full decile lower than their

brothers, despite being raised in the same household. This pattern mirrors the SES gradient observed in stock market participation and, together with the results in column (2), suggests that the larger gender gaps in financial outcomes among high-SES families have roots in earlier differences in math achievement formed during childhood.

This evidence aligns closely with findings from the education literature. Reardon et al. (2019) show that while boys and girls perform similarly in math on average, male-favoring gaps are larger in socioeconomically advantaged school districts, pointing to a role for gender stereotypes and parental resources in shaping math outcomes. Consistent with this view, Dossi et al. (2021) document that math gender gaps are most pronounced among children from affluent white families characterized by biased gender norms.

The top panel of Figure 1 illustrates this pattern using standardized test scores from the Stanford Education Data Archive (SEDA). We plot the distribution of gender gaps in math achievement for grades 3 through 8 across low- and high-SES Census tracts, where high-SES tracts are defined as those with average income and educational attainment above the median across all tracts in 2017, and remaining tracts are classified as low SES. The figure shows a clear relationship between socioeconomic advantage and larger male-favoring math gaps, based on test scores from millions of students, closely mirroring our within-family results in column (3).

Column (4) turns from math achievement to math attitudes in order to understand what drives the achievement gaps documented in column (3). Conditioning on math test performance, we find no meaningful gender difference in math attitudes among children from low-SES families. In high-SES families, however, a pronounced gap emerges: girls report significantly lower confidence in and enjoyment of math than their brothers, even when they perform at similar levels. This pattern indicates that gender differences in math attitudes are not simply a byproduct of achievement gaps, but instead may play a central role in their formation. In other words, even when girls in high-SES families demonstrate strong math performance, they report weaker identification with math, suggesting that attitudes constitute an independent margin along which gender differences emerge.³

Column (5) provides further evidence that math attitudes are an input into skill formation

³As additional evidence on gender differences in math attitudes and beliefs, we examine data from the 2012 U.S. PISA Student Questionnaire, which surveys 15-year-olds on beliefs, attitudes, and background characteristics. Following Breda et al. (2020), we construct a standardized measure of math beliefs that combines students' perceived control over math with parents' views on the importance of math for careers. Consistent with our results in column (4) of Table 3, gender differences in math beliefs are small among students from less advantaged backgrounds but widen with parental education and wealth, driven by stronger math beliefs among boys in more advantaged families. Appendix C describes the analysis and Figure C1 presents these patterns.

rather than a passive reflection of ability. Math achievement is highly persistent over time, but math attitudes also predict future math performance even after controlling for prior achievement. Children who report greater confidence and enjoyment of math subsequently improve their math performance relative to peers with similar initial test scores—a pattern documented by Susperreguy et al. (2018) across multiple data sources, and which we show holds within siblings raised in the same family. Taken together, the results in columns (4) and (5) point to a dynamic mechanism: gender differences in math attitudes, particularly in high-SES families, shape the evolution of math skills over time and contribute to the achievement gaps observed in later childhood and adolescence. This mechanism aligns closely with our adult results, where differences in math achievement and attitudes help explain gender gaps in stock market participation that are largest in socioeconomically advantaged environments.

2.5 Social Norms and Math Gender Stereotypes

Our findings connect closely to a growing literature showing that gender differences in math-related attitudes and outcomes are often most pronounced in more advantaged and ostensibly egalitarian settings. A key contribution is Breda et al. (2020), who link the “gender-equality paradox”—the tendency for gender segregation across fields and occupations to be greater in more developed and egalitarian countries (Charles and Bradley, 2002; Stoet and Geary, 2018)—to math-related gender stereotypes. They show that stereotypes associating math with males are stronger in these settings, even though more traditional gender-role beliefs, such as opposition to women’s higher education or support for women’s domestic roles, are less common. Their interpretation emphasizes that greater financial flexibility and parental involvement may allow these stereotypes to play a larger role in shaping educational and occupational choices.

An important implication of this work is that gendered beliefs about math need not stem from explicit barriers or overt discrimination, but may instead intensify precisely where access, resources, and formal equality are greatest. Our results provide micro-level evidence consistent with this view and extend it to financial outcomes. Within high-SES families, where resources and opportunities are abundant, gender differences in math attitudes emerge early, shape the accumulation of math skills, and are reflected in gender gaps in stock market participation later in life.

To further illustrate how math-related gender attitudes and skills manifest beyond childhood and adolescence, we develop a novel measure of gender differences in the math intensity of occupations at the Census-tract level. This measure combines information on the gender composition of occupations within each tract with occupation-level assessments of math skill

importance from the Occupational Information Network (O*NET). The resulting measure captures gender differences in the math intensity of occupations held by men and women within the same local labor market. As shown in the bottom panel of Figure 1, these occupational math skill gaps favor women in low-SES Census tracts but reverse sharply in high-SES tracts, closely mirroring the patterns observed earlier in childhood math achievement and attitudes and paralleling international evidence underlying the gender-equality paradox.

A large literature emphasizes that parents' beliefs, preferences, and skills shape both their children's educational trajectories and the broader environments in which children develop (e.g., Parsons et al., 1982; Eccles, 1994). Viewed through this lens, the gendered norms that influence children's math attitudes and achievement may also be reflected in adult occupational sorting within the same communities. Our findings are consistent with this interpretation: gender gaps in math-related skills among students and in the math intensity of occupations held by adults move in parallel across socioeconomic contexts, shifting from female-favoring to male-favoring as community income and education levels rise. Taken together, the evidence suggests that math-related gender attitudes differ systematically across environments and persist across childhood, adulthood, and the local labor market, with long-run implications for skill accumulation and financial outcomes such as stock market participation.

We next turn to mortgage choice and refinancing, another central household financial decision with substantial implications for long-run wealth accumulation. This setting provides a useful test of whether the patterns documented above reflect a broader household-finance mechanism rather than a friction specific to stock market participation. Mortgage decisions differ from equity market participation in institutional setting, but they similarly require households to recognize, evaluate, and act on financially valuable opportunities. In particular, choosing among mortgage alternatives and deciding whether to refinance involve complex, quantitatively intensive tradeoffs over borrowing costs, loan terms, and future interest-rate changes. If gendered math attitudes and quantitative skills shape financial behavior more broadly, similar gender-by-SES patterns should emerge in mortgage markets. In the following section, we examine whether the gender gaps we document in stock market participation extend to mortgage choice and refinancing behavior, and whether these gaps are likewise larger in socioeconomically advantaged environments.

3 Mortgage Outcomes

3.1 Mortgage Data

The Home Mortgage Disclosure Act (HMDA) requires most mortgage lenders to publicly report loan-level information on mortgage originations, including applicant demographics, income, loan characteristics, and property location, resulting in coverage of roughly 90 percent of U.S. mortgage originations. Following reporting expansions that took effect in January 2018, HMDA includes detailed measures of loan pricing (interest rates, points, fees, and lender credits), applicant credit risk (debt-to-income and loan-to-value ratios), loan purpose and term, occupancy status, and property characteristics, all of which are standard inputs in lenders’ underwriting decisions. Using this expanded dataset, we study conventional first-lien mortgage originations for single-family properties originated between January 2018 and December 2019, which allows us to observe these loans over several subsequent years.

While the expanded HMDA data provide detailed information on mortgage pricing and borrower characteristics, they do not include borrower credit scores or allow us to follow loans after origination. To incorporate these additional variables and outcomes, we merge HMDA originations with loan-level data from Fannie Mae and Freddie Mac, which include credit scores, more granular measures of borrower risk, precise origination timing, and monthly loan performance needed for refinancing and default analysis. The merge is conducted using a combination of borrower, loan, property, and geographic characteristics, and we retain only loans with a unique match across datasets. Details of the matching procedure are provided in Appendix Section B2. The resulting sample includes GSE-securitized loans originated in 2018 and 2019.

3.2 Additional Datasets

We obtain Census tract-level measures of median household income, educational attainment, and household composition from the American Community Survey (ACS) 5-year estimates ending in 2017. Educational attainment is measured as the share of adults aged 25 and older with a bachelor’s degree or higher. To characterize household composition among single adults, we compute the shares of single women and single men with children in each tract.

We draw standardized test score data from the Stanford Education Data Archive (SEDA; Reardon et al., 2021). Specifically, we use district-level gender gaps in math and reading achievement for grades 3 through 8, averaged across cohorts from 2009 to 2018. To link these measures to HMDA loan records, we map school districts to the Census tracts they serve.

We also use SEDA’s composite socioeconomic status measure, which captures average parental socioeconomic characteristics in public schools, and which Reardon et al. (2019) employ to study cross-sectional variation in gender achievement gaps across U.S. school districts.

We construct a novel tract-level measure of the gender gap in math skills, shown in the bottom panel of Figure 1, by combining occupation-level assessments of math importance from O*NET with local gender-specific employment shares. O*NET provides math skill importance scores from 1 to 5 for nearly one thousand occupations, which capture the extent to which mathematical abilities are required for job performance. We aggregate these to major occupational groups using national employment weights from the 2017 Bureau of Labor Statistics’ Occupational Employment and Wage Statistics and compute gender-specific weighted averages within each tract using male and female employment counts from the 2017 ACS 5-year estimates. The tract-level gender gap is defined as the difference between male and female average math skill importance. Results are robust to alternative O*NET measures, including math skill level, math knowledge level, and math knowledge importance. We apply the same occupation-based approach to construct gender gap measures for active listening, speaking, reading, and writing skills, which we use in robustness analyses. Details are provided in Appendix Section B3.

3.3 Summary Statistics

Table 4 summarizes the key variables for mortgages issued to single borrowers in our sample, while Table 5 reports statistics by borrower gender and by low- and high-SES Census tracts. High-SES tracts are defined as those with above-median income and educational attainment, where medians are computed across tracts and weighted by mortgage originations.⁴ Appendix Tables B2 and B3 present analogous splits by income and education separately; the high-SES indicator is strongly correlated with tract income and education, with pairwise correlations exceeding 70 percent (see Appendix Table B4). All continuous variables are winsorized at the 0.5 percent level.

Women account for 42 percent of single borrowers overall, and their presence is substantial across counties, as illustrated in Appendix Figure B1. Even at the 10th percentile of the county distribution (weighted by originations), women represent 29 percent of single borrowers; at the 90th percentile, the share rises to 48 percent (35 and 44 percent at the 25th and 75th percentiles, respectively). Low-SES tracts have an average income of \$58,000 and a college attainment rate of 27 percent, compared with \$103,000 and 54 percent in high-SES tracts.

⁴Appendix Table B1 provides detailed definitions of the main variables used throughout the mortgage analyses.

Gender gaps in student math test scores are reported as the difference between the average cohort-standardized male score and female score, expressed in standard deviations; for example, a gap of 0.5 indicates that the average male student scored half a standard deviation higher than the average female student in the same cohort. Average gender gaps in math scores are small but vary systematically with tract SES. In low-SES tracts, the average gap is -0.01 standard deviations, favoring females, while in high-SES tracts it reverses to 0.01 standard deviations, favoring males. A similar pattern emerges for occupational math skills: women exhibit a 0.11 standard deviation advantage in low-SES tracts, which reverses to a 0.07 standard deviation advantage for men in high-SES tracts. As shown in Figure 1 and discussed in Sections 2.4 and 2.5, the distributions of these gaps differ markedly across socioeconomic environments, highlighting pronounced gender differences by tract SES.

The average mortgage rate in our sample is 4.4%, and the average rate spread, which is the difference between the covered loan’s APR and the average prime offer rate for a comparable transaction as of the date the interest rate is set, is 0.5%. Mortgage rates in low-SES tracts exceed those in high-SES tracts by approximately 15 basis points, and rate spreads are higher by about 20 basis points. We observe gender gaps in raw averages, which are more pronounced in high-SES tracts. Average mortgage rates for females are 3 basis points higher than for males in low-SES tracts, and 6 basis points higher in high-SES tracts. Gender differences in rate spreads are smaller, with a difference of 3 basis points in high-SES tracts and 1 basis point in low-SES tracts (rounded to two significant digits). Loan costs average 1.9% of the loan amount, which are approximately 0.7% higher in low-SES neighborhoods. Females pay 13 basis points more in net loan costs compared to males in low-SES neighborhoods, with this spread rising to 20 basis points in high-SES neighborhoods.

There is substantial heterogeneity in mortgage terms within both low- and high-SES tracts. Dispersion is similar across socioeconomic environments for mortgage rates and lender credits, while rate spreads, net loan costs, and upfront fees are noticeably more dispersed in low-SES tracts. This pattern holds whether dispersion is measured using standard deviations or interquartile ranges. Appendix Table B5 reports the full distribution of these variables by tract SES.

The average credit score of borrowers is 748. The difference in credit scores between borrowers in low-SES and high-SES neighborhoods is small, about 9 points and there is almost no difference in the credit scores of male versus female borrowers. In both low- and high-SES tracts, male borrowers have higher incomes and loan amounts, slightly higher loan-to-value ratios, and lower debt-to-income ratios compared to female borrowers.

White borrowers make up approximately 70% of our sample.⁵ The share of White borrowers is similar for men and women in low-SES tracts (about 69.5 percent), but differs in high-SES tracts, where it is higher among women (72 percent) than among men (67.5 percent).

To understand the household composition of single borrowers, specifically whether they have children, we supplement the loan-level data with ACS data on single-adult households in the relevant Census tracts, as household composition details aren’t available at the loan level. According to these combined data, 51.5% of households led by single females have children, with this proportion slightly decreasing in more affluent neighborhoods. Among single-male-led households, 47.6% have children, and this share increases slightly in more affluent neighborhoods. We revisit these statistics in our supplementary analyses and robustness checks.

Appendix Tables B6 and B7 expand upon Tables 4 and 5 by summarizing key variables for all mortgages in the HMDA-GSE merged dataset, including both single and joint mortgages. Both tables distinguish single female borrowers, single male borrowers, female primary borrowers in joint mortgages, and male primary borrowers in joint mortgages. In the full sample reported in Appendix Table B6, these groups account for 23.2%, 32.2%, 11.7%, and 32.9% of loans, respectively. The primary borrower is typically the individual who completes and submits the mortgage application, likely leading decision-making throughout the process.

While our analysis primarily focuses on single borrowers, descriptive comparisons of joint applications led by male versus female primary borrowers reveal directionally similar patterns. Among joint mortgages, the average mortgage rate for female primary borrowers is 4 basis points higher than for male primary borrowers, and the raw gap in the rate spread is 5 basis points, compared to 2 basis points for single borrowers. Joint mortgages also feature higher loan amounts and lower debt-to-income ratios than those of single borrowers. These patterns are consistent with the gaps we document among single borrowers, but they should be interpreted as descriptive and suggestive: the designation of a “primary” borrower in joint applications is unlikely to be random and may reflect differences in income, credit strength, household roles, or selection into being listed first.

⁵The “White” category includes borrowers with “White” as their reported race in HMDA, excluding those who indicated an ethnicity of “Hispanic or Latino.” The remaining racial and ethnic breakdowns are Hispanic or Latino (8.7%), Asian (7.7%), Black (5.7%), American Indian (0.5%), Hawaiian (0.3%), Two or more minority races (0.1%), and Race not available (7.4%).

3.4 Gender Gaps in Mortgage Costs and Local Socioeconomic Factors

To estimate gender gaps in mortgage costs and examine how they vary with local socioeconomic conditions, we use ordinary least squares regressions of various mortgage cost measures on an indicator for female borrowers and its interaction with SES. Our rich loan-level data allow us to include Census tract $\times$ origination year-by-month fixed effects, so identification comes from comparisons among borrowers within the same Census tract at the same point in time. We additionally control for a detailed set of borrower, loan, and property characteristics, described below, to further isolate the role of gender and its interaction with local socioeconomic status. We estimate the following regression:

$$Y_{i,j,c,k,t} = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Female}_i \times \text{SES}_c + \Gamma Z_{i,j} + \alpha_c \times \alpha_t + \alpha_k + \epsilon_{i,j,c,k,t}, \quad (3)$$

where $Y_{i,j,c,t}$ represents the outcome of interest for borrower i , with loan characteristics j , in Census tract c , lender k and loan origination month t . Our primary outcomes are the mortgage interest rate and net loan costs, defined as total points and fees minus lender credits, expressed as a fraction of the loan amount at origination. We also examine the rate spread reported in HMDA, defined as the difference between the loan’s annual percentage rate (APR) and the average prime offer rate for a comparable transaction on the date the rate is set, as well as total points and fees and total lender credits, each measured as a fraction of the loan amount.

Female_i is an indicator variable that takes the value of 1 if the borrower is female and 0 if the borrower is male. The socioeconomic status SES_c is a dummy variable equal to 1 if the Census tract’s median income and educational attainment exceed the sample median, computed across Census tracts and weighted by mortgage originations.

$\Gamma Z_{i,j}$ denotes a rich set of borrower and loan characteristics that may affect mortgage pricing. The pricing controls follow the loan-level price adjustment grids of Fannie Mae and Freddie Mac, which determine the credit-risk adjustments to the guarantee fees paid by lenders. Specifically, we include dummies for nine loan-to-value (LTV) categories, nine credit score categories, manufactured homes, investment properties, high-balance conforming loans, subordinate liens, loan purpose (home purchase, refinance, or cash-out refinance), and the number of units (one to four). As argued by Bartlett et al. (2022), these pricing grids capture the legitimate credit-risk determinants of GSE pricing, so any residual pricing differences beyond these controls are unlikely to reflect observable credit risk.

Additional controls include dummies for eight borrower race and ethnicity categories, income

deciles, loan amount deciles, three loan term intervals, and seven borrower age intervals. $\alpha_c \times \alpha_t$ denotes Census tract $\times$ origination year-month fixed effects, and α_k denotes lender fixed effects. Standard errors are clustered at the Census tract level. Double clustering at the Census tract and origination year-month level yields similar results.

Table 6 presents the results of the estimation of equation (3). The columns progressively include more controls for each mortgage outcome. The top panel presents results for the mortgage interest rate as the dependent variable, while the bottom panel reports results for net loan cost at origination.

The dependent variable in the top panel of Table 6 is the mortgage interest rate. Columns (1), (4), and (7) report estimates of the unconditional gender gap prior to the inclusion of SES interactions. Column (1) includes no borrower or loan controls, column (4) adds pricing-related variables, and column (7) incorporates the full set of borrower and loan characteristics available in the data.

Within the same Census tract and origination month, mortgages issued to single female borrowers carry interest rates that are 2.8 basis points higher than those issued to single male borrowers. This gap widens to 4.2 basis points when pricing controls such as loan-to-value ratios and credit scores are included, indicating that these variables do not explain the disparity and, if anything, partially mask it. For context, Bartlett et al. (2022) document racial disparities in mortgage rates of roughly 2 basis points for Black and Hispanic borrowers on purchase GSE loans and 0.8 basis points on refinance GSE loans using a similar pricing framework. When we further control for borrower race, income, age, and loan amount, the gender gap in mortgage rates declines to 0.9 basis points, suggesting that these factors account for part, but not all of the observed difference.

Columns (2), (5), and (8) extend the specification by interacting borrower gender with an indicator for high socioeconomic status of the Census tract, using progressively richer control sets. Results are very similar when the female indicator is interacted separately with tract income and tract education, as reported in Appendix Table B9), which motivates our use of a combined SES indicator in the main analysis. Columns (3), (6), and (9) further include lender fixed effects.

The estimates show that gender gaps in mortgage interest rates roughly double in high-SES Census tracts, mirroring the patterns observed for stock market participation. Interestingly, this pattern contrasts with Bosshardt et al. (2026), who find that gender gaps in home appraisals, an outcome largely determined by third-party assessments rather than borrower search or financial decision-making, are more pronounced in lower-SES neighborhoods. Including lender fixed

effects has little effect on the results, indicating that differential sorting across lenders does not explain the observed gaps. This is notable because more than half of the lenders in our sample originate only one mortgage per Census tract per month; including lender fixed effects therefore effectively restricts the sample to larger, more established lenders. The persistence of gender gaps within this subset suggests that the disparities are widespread rather than driven by a small set of lenders.

The bottom panel of Table 6 reports estimates for net loan costs, and shows patterns similar to those for mortgage interest rates. In baseline specifications, mortgages issued to single female borrowers carry net loan costs that are 17 basis points higher than those for single male borrowers. This gap narrows to 12 basis points after adding pricing controls and reverses to -0.5 basis points with the inclusion of the full set of borrower and loan characteristics. However, the interaction between the female indicator and high-SES remains positive across all specifications. Even in the fully controlled model, while the gender gap in net loan costs is -0.7 basis points in low-SES tracts, it reverses to 0.7 basis points in high-SES tracts, indicating higher borrowing costs for women in more advantaged areas.

The origination-pricing estimates are economically modest in the most saturated specifications, especially after conditioning on borrower income, loan amount, and other non-pricing characteristics. This is perhaps unsurprising in our setting: the loans we study are conventional conforming mortgages securitized by Fannie Mae and Freddie Mac, a highly standardized and regulated segment of the mortgage market in which pricing is closely tied to observable risk characteristics. The main takeaway is therefore not any single pricing estimate, but the consistency of the pattern: gender gaps widen with local socioeconomic status across a broad set of mortgage outcomes and specifications.

We conduct a series of robustness checks that examine additional outcome variables, alternative measures of local socioeconomic conditions, and key subsamples, including White borrowers, different age cohorts, and purchase versus refinance loans. All robustness results are reported in the Appendix.

We first consider additional mortgage pricing outcomes. Appendix Table B8 reports results for the rate spread and for the two components of net loan costs, upfront fees and lender credits. The rate spread results closely mirror those for mortgage interest rates, reinforcing our main findings. Decomposing net loan costs shows that single female borrowers pay slightly lower upfront fees than single male borrowers in low-SES tracts, but this pattern reverses in high-SES tracts. At the same time, women receive smaller lender credits, particularly in high-SES tracts, even after controlling for a rich set of borrower and loan characteristics. Taken together,

the additional pricing outcomes show that the SES gradient is not specific to mortgage rates, but also appears in rate spreads, points and fees, and lender credits.

We also assess robustness to alternative measures of local socioeconomic conditions. In addition to the separate income and education specifications discussed above, Appendix Table B9 reports results using a composite socioeconomic status index constructed by Reardon et al. (2019), which incorporates poverty and unemployment rates, food stamp participation, the prevalence of single-mother households, in addition to income and education. We also use the ratio of Census tract median family income to Metropolitan Statistical Area (MSA) median family income to capture relative affluence within metropolitan areas. Across all measures, gender gaps in mortgage costs are consistently larger in more socioeconomically advantaged areas.

Finally, the results are robust across key subsamples. Restricting the sample to White borrowers, who account for approximately 70 percent of single borrowers, yields similar patterns, alleviating concerns that the results are driven by correlations between gender and race (Appendix Table B10). The findings are also stable across age cohorts: Appendix Table B11 shows that larger gender gaps in high-SES areas persist across all ten-year age groups. In addition, Appendix Tables B12 and B13 demonstrate that the SES gradient in gender gaps holds separately for purchase and refinance loans.

3.5 Gender Gaps in Default

Could the gender gaps in mortgage costs documented in Section 3.4 reflect unobserved differences in default risk not captured by our pricing controls or borrower and loan characteristics? Prior evidence suggests that this explanation is unlikely. Goodman et al. (2016) document lower mortgage delinquency rates among female borrowers, while Fang and Munneke (2020) show that female borrowers paid higher mortgage rates even after explicitly accounting for predicted default and prepayment risk. From the lender’s perspective, differential default risk is also unlikely to explain our results because the mortgages in our sample are securitized and sold to investors, and from the investors’ standpoint, default risk is largely eliminated by GSE guarantees. Moreover, our pricing controls explicitly account for the GSE pricing grids used to set guarantee fees. As a result, neither lenders nor MBS investors are exposed to differential default risk across male and female borrowers.

We therefore examine whether the GSEs themselves face higher default risk from female borrowers. To do so, we estimate a monthly loan-level hazard specification in which the outcome is

an indicator for becoming 90 days or more past due. The specification uses the same borrower, loan, and timing controls that we later use to analyze prepayment behavior. To reduce computational burden, we analyze a 15% random sample of loans from the merged HMDA–GSE dataset. The average monthly default rate in our sample is 0.22 percent, and Table 7 reports the results.

We find that female borrowers have a significantly lower default hazard than male borrowers, indicating that lending to female borrowers is, if anything, less risky. The estimated default gap is economically meaningful, amounting to roughly 21 percent of the average monthly default rate among single borrowers. This estimate is stable across specifications and remains largely unchanged when controlling for mortgage pricing, loan characteristics, and borrower attributes. These findings are consistent with Gerardi et al. (2023), who document lower default rates among female borrowers in GSE loans, and with Delis et al. (2022), who find that women are less likely to default on small business loans.

Finally, we find no evidence that the default gap varies with local socioeconomic conditions: interactions between the female indicator and tract SES are small and statistically insignificant. Default risk therefore cannot explain either the level of the gender gap in mortgage costs or its larger magnitude in more socioeconomically advantaged areas.

3.6 Gender Gaps in Refinancing

Thus far, we have focused on gender differences in mortgage terms at origination for conventional loans securitized by Fannie Mae and Freddie Mac. We document statistically significant gaps, particularly in more affluent areas, even after controlling for a rich set of borrower, loan, and lender characteristics. While these effects are modest in absolute terms, reflecting the high degree of standardization in this market, they are comparable in magnitude to racial gaps documented in the literature.

Mortgage decisions, however, extend beyond loan selection at origination. A central post-origination decision is whether and when to refinance. Refinancing allows borrowers to benefit from declines in mortgage rates, and prior work shows that this channel has meaningful effects on household cash flows and monetary-policy transmission (Di Maggio et al., 2020). Descriptive evidence suggests that gender differences are particularly pronounced during refinancing booms: Goodman and Zhu (2020) show that while the share of home purchase loans to single female borrowers remained relatively stable, their share of refinancing loans declined markedly during the low-interest-rate period of 2020–2021, when the gains from refinancing were especially

large. Households that fail to refinance during such periods forgo these gains. We therefore next examine prepayment behavior to assess whether gender gaps also arise in refinancing decisions.⁶

Our analysis uses the same monthly loan-level hazard framework introduced in the default analysis, now replacing the default indicator with an indicator for voluntary prepayment. A loan is classified as prepaid if it is voluntarily paid off in full prior to maturity. The average monthly prepayment rate in our sample is 2.08 percent, computed from pooled loan-month observations spanning January 2018 to September 2023, where each active loan contributes one observation per month and the prepayment indicator equals one only in the final month of the loan.

The 2020–2021 period provides a particularly useful setting for studying refinancing behavior. Mortgage rates fell to historically low levels during this window, generating large potential savings for existing borrowers and triggering a broad refinancing wave. Figure 2 presents Kaplan–Meier estimates of prepayment hazard rates by gender for low- and high-SES Census tracts. The top panel plots the hazard as a function of months since origination, while the bottom panel plots it over calendar time. These estimates reflect unconditional average monthly prepayment rates, calculated as the share of loans prepaid in a given month among loans that reached that month without prior termination or censoring.

Comparing the top-left and top-right panels reveals several patterns. Borrowers in low-SES tracts refinance less frequently and tend to refinance later than those in high-SES tracts. During the peak refinancing waves—approximately 15 to 30 months after origination—the difference in refinancing rates between the two groups exceeds 1 percentage point. Gender differences also vary sharply by socioeconomic status. In low-SES tracts, refinancing rates are similar for men and women, whereas in high-SES tracts a gender gap of roughly 0.5 percentage points emerges and reaches nearly 1 percentage point at the early peak of the refinancing boom.

The bottom panels present the same comparisons over calendar time. Notably, refinancing among male borrowers in high-SES tracts peaked at over 6 percent in March 2021, while refinancing among female borrowers in those tracts remained closer to 5 percent. In contrast, refinancing rates for both men and women in low-SES tracts hovered around 4 percent throughout this period.

To examine how gender and local socioeconomic conditions relate to mortgage prepayment,

⁶Our data do not allow us to distinguish prepayments due to refinancing from those due to home sales. To the extent that single men and women differ in their holding periods, some differences in prepayment may reflect sales rather than refinancing. For example, Goldsmith-Pinkham and Shue (2023) report an average holding period of 5.34 years for single men and 5.76 years for single women, implying a slight difference in sales hazards.

we estimate the following linear probability model:

$$\text{Prepayment}_{i,j,c,t,\tau} = \beta_1 \text{Female}_i + \beta_2 \text{Female}_i \times \text{SES}_c + \Gamma Z_{i,j,t,\tau} + a_d + a_t + \epsilon_{i,j,c,t}, \quad (4)$$

where $\text{Prepayment}_{i,j,c,t,\tau}$ equals one if a mortgage for borrower i , with loan characteristics j , in Census tract c , originated in month t , is voluntarily prepaid in full in month τ , and zero otherwise. Once a loan is prepaid, it exits the sample. The control vector $Z_{i,j,t,\tau}$ includes all borrower and loan characteristics from equation (3), as well as property value at origination and loan age in months since origination. The specification includes county fixed effects (a_d) and origination year-month fixed effects (a_t). The unit of analysis is the loan-month, and standard errors are double clustered at the county and origination year-month levels.

Table 8 reports the results from estimating equation (4). Column (1) presents a specification that includes origination year-month and county fixed effects but excludes additional borrower and loan-level controls. Female borrowers refinance at a rate 23 basis points lower than male borrowers, a gap that exceeds 10 percent of the average monthly refinancing rate among single borrowers. Column (2) introduces an interaction with the high-SES indicator and shows that the gender gap more than doubles in high-SES Census tracts.

Columns (1) and (2) examine overall prepayment behavior without distinguishing whether refinancing is financially optimal. We next introduce a “rate drop” indicator to identify periods when refinancing is likely advantageous—specifically, when the prevailing market mortgage rate falls below the borrower’s origination rate. The indicator equals one if the 30-year fixed mortgage rate at origination, measured using the Freddie Mac Primary Mortgage Market Survey (PMMS), exceeds the market rate in a given month.

Column (3) interacts the rate drop indicator with borrower gender and tract SES. The estimates show that gender gaps in refinancing are concentrated during periods of rate declines and are especially pronounced in high-SES areas. Notably, in column (9), which includes the full set of controls, women are more likely than men to prepay outside of rate-drop periods, particularly in high-SES tracts. In contrast, during rate-drop periods, when refinancing is most beneficial, female borrowers in high-SES areas exhibit a monthly prepayment hazard that is approximately 0.3 percentage points lower than that of male borrowers, implying substantial missed savings from delayed or foregone refinancing.

Appendix Table B14 shows that the refinancing results are robust to alternative measures of local socioeconomic conditions. We further assess the robustness of the results to an alternative definition of refinancing opportunities. Specifically, we construct an alternative rate-drop indicator that equals one when the contract mortgage rate exceeds the prevailing market rate

in a given month. The results from this robustness check are almost identical to our baseline findings.

3.7 Gender Gaps in Overall Mortgage Costs

Thus far, we have shown that single female borrowers receive less favorable mortgage terms than single male borrowers, with these disparities widening in high-SES areas. We also find that female borrowers are less likely to refinance, particularly during periods of interest rate declines, and that this gender gap is likewise larger in more affluent regions. When borrowers who face less favorable terms at origination are also less likely to refinance, the combined cost of higher borrowing rates and missed refinancing opportunities is likely to exceed the impact of either margin in isolation.

To assess this combined effect, we link the initial mortgage terms to prepayment behavior and track each loan until it closes (due to prepayment or default) or until September 2023, whichever occurs first. Using this data, we calculate a hypothetical 10-year overall mortgage cost, which includes the origination costs and interest payments on the original loan amount over a 10-year period. If the original loan is not prepaid by September 2023, we assume it continues for the full 10 years. For loans that are paid off before September 2023, we estimate a new interest rate by subtracting the difference in market rates between the origination and prepayment month from the original rate, and we assume the borrower incurs refinancing costs equal to the original loan costs (fees minus lender credits). These are conservative assumptions: borrowers may refinance multiple times, especially those who refinance early, and subsequent origination costs are likely to be lower than those of the initial loan.⁷ Given the typically small principal amortization in the early years of a long-term mortgage, we assume the loan balance remains unchanged.

We find that 68% of mortgages in our sample were paid off by September 2023, largely driven by the historically low mortgage rates following the COVID pandemic. On average, market interest rates fell by 0.81% from origination to prepayment for the mortgages in our sample, though the benefits of refinancing were partially offset by incurring new origination costs. Table 9 presents the results of estimating equation (3) using overall mortgage cost, expressed as a percentage of the loan amount, as the dependent variable.

⁷In the regressions reported in Table 6, we estimate that net origination costs for refinance loans are 0.26 percentage points lower than those for purchase loans, while cash-out refinances are 0.22 percentage points more expensive, after controlling for loan and borrower characteristics relevant to risk-based pricing. These results indicate that the origination costs for pure refinance loans are substantially lower than the unconditional average of 1.93% reported in Table 4.

Our estimates imply that female borrowers incur higher overall mortgage costs over the first ten years of the loan. Relative to single men borrowing in the same tract and time period, women pay approximately 0.6 percent more of the loan value in low-SES tracts and more than 1 percent more in high-SES tracts. These disparities are largely unchanged when controlling for risk-based pricing factors, though they are attenuated once additional non-pricing borrower and loan characteristics are included. Appendix Table B15 confirms that, like earlier results on mortgage costs and refinancing, overall cost results are robust to alternative SES measures.

In dollar terms, these differences correspond to roughly \$3,000 in additional payments for women on a \$300,000 mortgage in high-SES areas and about \$1,200 on a typical \$200,000 mortgage in low-SES areas. The gaps are even larger at the upper end of the socioeconomic distribution: at the 99th percentile of tract income and education, excess costs reach approximately 1.5 percent of a \$400,000 loan—about \$6,000. Even after conditioning on all observable characteristics, which eliminates the gap in low-SES tracts, women in high-SES areas still pay about 0.2 percent more than otherwise comparable male borrowers.

3.8 Inspecting the Mechanism

The results thus far show that the socioeconomic gradient in gender gaps is not confined to stock market participation, but also appears in mortgage pricing and refinancing. Women—particularly those living in more affluent, higher-income, and higher-education communities—borrow on less favorable terms at origination and are also less likely to refinance when interest rates decline. This recurrence across domains is central to our interpretation: it points away from a market-specific explanation and toward a broader household-finance mechanism.

In the mortgage market specifically, these gaps may reflect less effective mortgage shopping, reduced search effort, or difficulties in evaluating loan terms that require comparing rates, fees, and future payment streams. For example, borrowers may focus primarily on the headline interest rate, which is more salient, while overlooking fees or the trade-offs between rates and points. These challenges are compounded by refinancing decisions, which, like stock market participation, require proactive engagement and careful quantitative evaluation of intertemporal costs and benefits. Difficulties in processing this information can lead both to higher borrowing costs at origination and to missed opportunities to refinance when rates fall.

3.8.1 Gender Gaps in Financial Literacy

A large literature documents that women, on average, exhibit lower levels of financial literacy and different search behaviors than men (Lusardi and Mitchell, 2014; Fonseca et al., 2012; Malliaris et al., 2022). Consistent with this evidence, our stock market participation results in Table 2 show that financial literacy explains a substantial share of both the level of the gender participation gap and its increase in high-SES areas. To further examine how financial literacy interacts with gender and socioeconomic conditions in the mortgage market, and how these factors shape mortgage search behavior, we turn to data from the National Survey of Mortgage Originations (NSMO).

The NSMO is a nationally representative survey of newly originated residential mortgages in the United States, jointly conducted by the Federal Housing Finance Agency (FHFA) and the Consumer Financial Protection Bureau (CFPB). It provides detailed information on borrowers' experiences during the mortgage process, their perceptions and understanding of mortgage products, and links these responses to administrative data on borrower and loan characteristics. Because the public-use NSMO does not report borrowers' geography, we use it for the borrower-level measures it uniquely provides, mortgage knowledge and shopping behavior, while relying on the administrative mortgage data for pricing and refinancing outcomes.

We begin by assessing how knowledgeable borrowers are about the mortgage process and the important economic decisions they made, by evaluating their answers to the following questions: "How well could you explain to someone the... Process of taking out a mortgage / Difference between a fixed- and an adjustable-rate mortgage / Difference between a prime and subprime loan / Difference between a mortgage's interest rate and its APR / Amortization of a loan / Consequences of not making required mortgage payments / Relationship between discount points and interest rate." Respondents chose from a three-point scale: "Not at all" (coded as 1) to "Very" (coded as 3). Section D1 in the Appendix provides detailed information on the survey questions and the distribution of responses.

While each question addresses an important feature of mortgages, borrowers find some questions easier than others. For example, fewer than 10% of borrowers report difficulty explaining the process of taking out a mortgage, the difference between fixed- and adjustable-rate mortgages, and the consequences of not making required mortgage payments. In contrast, proficiency is lower and responses are more dispersed for questions involving amortization, the distinction between the interest rate and APR, and the relationship between discount points and the interest rate. These latter concepts require a deeper understanding of mathematical relationships and trade-offs, and the clear split in responses motivates our classification of the

first three questions as measuring “Basic Literacy” and the remaining four as measuring “Advanced Literacy.” This distinction follows van Rooij et al. (2011), who similarly separate basic and advanced financial knowledge and show that advanced literacy is particularly important for complex financial decisions such as stock market participation.

These multi-item mortgage-literacy measures allow us to move beyond the single compound-interest item available in the PSID and examine whether gender gaps are concentrated in more quantitatively demanding financial concepts. As with the canonical financial literacy questions, these measures are knowledge-based: they assess borrowers’ understanding of mortgage concepts rather than their actions during the mortgage process. We examine shopping behavior separately in Section 3.8.2, where we study its relationship to mortgage literacy.

To estimate how mortgage literacy is related to gender and socioeconomic status, we estimate the following regression:

$$L_{i,t,s} = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{SES}_i + \beta_3 \text{Female}_i \times \text{SES}_i + \Gamma Z_i + a_t + \delta_s + \epsilon_{i,t,s}, \quad (5)$$

where $L_{i,t,s}$ denotes the mortgage literacy level of borrower i who originated a loan in month t and was surveyed in wave s . Female_i is an indicator equal to one if the borrower is female, and zero otherwise. SES_i is defined analogously to our earlier analysis and equals one if borrower i has above-median income and has completed college education. Appendix Table D1 provides detailed definitions of the main NSMO-based variables used in the analysis.

To isolate the effects of gender and socioeconomic status, we control for a comprehensive set of borrower characteristics, denoted by ΓZ_i . These controls include indicators for five income categories, four education levels, five race and ethnicity groups, six age groups, and nine credit score categories. All specifications include origination year-month fixed effects a_t and survey wave fixed effects δ_s . We apply the NSMO-provided survey weights, which account for sampling design and nonresponse. The sample is restricted to single borrowers with conventional loans, ensuring consistency with the analyses throughout the paper.

Table 10 reports coefficient estimates from equation (5) and confirms several patterns documented in the financial literacy literature. Female borrowers exhibit lower mortgage literacy across all questions, with substantially larger gaps for more advanced concepts. Mortgage literacy also increases with socioeconomic status. Because the regressions include detailed controls for income and education, the SES coefficient captures the incremental effect of being both high-income and highly educated, rather than the additive effects of either characteristic alone.

The table further shows that gender gaps in mortgage literacy vary sharply with both

socioeconomic status and the complexity of financial knowledge. Among low-SES borrowers, women score lower than men on advanced concepts such as APR, amortization, points, and subprime distinctions, while gender differences in basic mortgage knowledge are relatively small. High socioeconomic status is strongly associated with higher advanced mortgage literacy for men, but this gradient is much weaker for women. As a result, gender gaps widen substantially in high-SES groups for advanced literacy measures, with female-SES interaction effects ranging from -0.12 to -0.18 standard deviations. In contrast, gender differences in basic mortgage literacy exhibit little to no socioeconomic gradient.

These patterns indicate that gender gaps in mortgage knowledge are concentrated in complex, quantitatively intensive concepts and expand further in socioeconomically advantaged environments, even after accounting for income and education. While prior work has documented overall gender gaps in financial literacy, attention to how these gaps vary with socioeconomic status has been limited. For example, Fonseca et al. (2012) find an interaction between gender and education, though not income, and do not emphasize this dimension; much of the existing literature focuses primarily on basic literacy measures. Our results highlight that both the depth of financial knowledge and its socioeconomic context are central to understanding gender differences.

Appendix Figure D1 illustrates advanced literacy by income and education. Advanced literacy increases with income for both men and women, but the gender gap widens noticeably at higher income levels. The heterogeneity in education gradient is even more pronounced: literacy rises steadily with education for men, whereas for women it plateaus after some college and slightly declines at the postgraduate level, resulting in a large gender gap among highly educated borrowers.

We examine whether the results are sensitive to alternative measurement and sample definitions. First, unlike HMDA, the NSMO includes information on whether borrowers are unmarried and report no spouse or partner sharing household decision-making responsibilities. We use this information to further restrict the single-borrower sample. This restriction retains roughly half of the NSMO single-borrower sample and provides a useful check on whether the results are sensitive to the presence of a spouse or partner outside the mortgage record. Second, we construct a borrower sophistication index following Bhutta et al. (2026), using the closest analogue that can be constructed from the variables available in the public NSMO data. This alternative measure draws on a different set of NSMO questions and complements our mortgage-literacy measures by providing an independently established way to capture borrower sophistication during the mortgage process. Appendix Table D2 reproduces the mortgage-literacy analysis in Table 10 for single borrowers with no reported spouse or partner, while Appendix Table

D3 reports results for the borrower sophistication index. The results are similar across these alternative specifications.

A potential concern with self-reported literacy measures is that they may partly reflect differences in confidence rather than underlying knowledge. Prior research shows that men are more likely to overstate and women to understate their performance in math-related settings (Barber and Odean, 2001; Reuben et al., 2014), contributing to baseline gender gaps in financial literacy (Bucher-Koenen et al., 2017). However, confidence alone is unlikely to explain the socioeconomic gradient we document: it would require gender differences in over- or underconfidence to increase systematically with income and education. Moreover, similar socioeconomic gradients appear in our earlier analyses of standardized math test scores and occupational math intensity, which are not self-reported and therefore not susceptible to confidence bias. The consistency of these patterns across independent measures supports the interpretation that differences in quantitative skills contribute to the widening gender gaps in mortgage borrowing costs and refinancing behavior observed in more socioeconomically advantaged communities.

3.8.2 Mortgage Shopping Behavior

Next, we examine gender differences in mortgage shopping behavior and their relationship to financial literacy as measured in the NSMO. We focus on several key stages of the process: borrowers' initial familiarity with mortgage concepts, whether they seriously considered more than one lender, whether they were presented with multiple loan types or features, and whether they verified the offered terms using outside sources. Section D1 in the Appendix provides detailed information on the survey questions used to assess shopping behavior.

Table 11 reports the results. At each stage, female borrowers engage in less intensive search than male borrowers. They begin the process with lower self-reported familiarity, consider fewer lenders, review fewer loan options, and are less likely to confirm terms through additional sources. These gaps are closely related to financial literacy. When we include our literacy measure in the regressions, the coefficient on the female indicator declines substantially for most search outcomes, while the literacy coefficient remains stable. This pattern suggests that lower financial literacy among women is an important contributor to weaker mortgage shopping behavior. Appendix Table D4 shows that these findings are robust to restricting the sample to single borrowers with no reported spouse or partner.

Appendix Table D5 further shows that the socioeconomic gradient in initial familiarity with mortgage concepts differs sharply by gender. The gender gap in familiarity with interest rates, mortgage types, and down payments widens with income and education, reinforcing the broader

pattern that literacy and search gaps are most pronounced in socioeconomically advantaged environments.

Administrative application data provide a complementary, though less straightforward, perspective on shopping behavior. In HMDA, applications can be originated, denied, withdrawn, or not accepted, and borrower-initiated withdrawals are often interpreted as evidence of shopping (Jiang et al., 2024). In our data, presented in Appendix Table B17, withdrawal rates are slightly higher for men than for women, particularly in high-income, high-education tracts. However, denial rates vary by more than 10 percentage points between low- and high-SES areas, whereas withdrawal rates differ by only about 1 percentage point. This contrast suggests that the motivation for multiple applications may differ across socioeconomic environments: in lower-SES areas, borrowers may apply to multiple lenders to improve approval odds, while in higher-SES areas withdrawals are more likely to reflect active search for better terms. As a result, application outcomes alone provide an imperfect proxy for mortgage shopping intensity.

While documented differences in search behavior are likely contributors to the gender gaps, they are unlikely to be the only factors. Malliaris et al. (2022), using NSMO data, finds that performing a competitive search is not a substitute for financial sophistication. Considering the nature of the advanced literacy questions we examine and the differences we document, this finding is not surprising. For example, fewer than 30% of borrowers in the NSMO dataset state that they know the relationship between discount points and interest rates well, and a third of them state that they do not know it at all. Even if potential borrowers perform a thorough search and compare different loan options, they can make suboptimal choices due to not fully understanding the trade-offs inherent in mortgage options. Therefore, gender gaps in financial literacy, and the conceptual understanding of mathematical relationships it requires, are likely to contribute to gender gaps in mortgage outcomes through multiple channels, including less effective search and difficulty evaluating complex mortgage tradeoffs.

3.8.3 Math Skills and Gender Gaps in Mortgage Outcomes

A substantial body of prior work links math education and numeracy to financial outcomes. Consistent with this evidence, our earlier results show that larger gender differences in math attitudes and achievement within high-SES families account for a meaningful share of the higher gender gap in stock market participation observed in these families. Together, these findings make it plausible that the larger gender gaps in math attitudes and skills observed in more affluent areas also contribute to the relationship between socioeconomic status, measured by income and education, and gender disparities in mortgage outcomes. To examine this channel

directly, we replicate our earlier analysis of the overall mortgage cost replacing the high-SES indicator with measures of local gender gaps in math skills.

To connect the SES gradient in mortgage costs to local math-related gender gaps, we return to the overall mortgage cost specification from Section 3.7, which combines costs at origination with subsequent prepayment behavior. Columns (1), (4), and (7) of Table 9 show that gender gaps in overall costs are larger in high-SES areas. In the remaining columns, we replace the Female $\times$ SES interaction with interactions between the female indicator and local measures of gender gaps in math skills, specifically student math achievement gaps and occupational math-intensity gaps. The estimates show that the female cost gap is larger in areas with more male-favoring math gaps, with magnitudes comparable to those associated with high-SES areas.⁸

Could the observed gender gaps in mortgage outcomes reflect broader skill differences rather than gaps in math skills specifically? Our earlier results on stock market participation suggest otherwise: column (2) of Table 3 shows that reading achievement has no explanatory power once math achievement is controlled for. We test this hypothesis directly in the mortgage context through a series of robustness tests. In Appendix Table B16, we include local gender gaps in student reading achievement alongside math achievement, and gender gaps in basic occupational skills from O*NET, specifically active listening, speaking, reading, and writing alongside occupational math skills. Across both sets of tests, these placebo measures have little explanatory power once gender gaps in math skills are included. These findings mirror our stock market results and reinforce the interpretation that math skills, rather than general cognitive or communication skills, play a distinct role in shaping gender disparities in mortgage outcomes. These findings mirror our stock market results and reinforce the interpretation that math skills, rather than general cognitive or communication skills, play a distinct role in shaping gender disparities in mortgage outcomes. They also support the broader view that math-related gender norms may help explain cross-sectional variation in these gaps.

3.8.4 Alternative Explanations

In addition to gender differences in financial literacy and the persistent stereotype that “math is not for girls,” we consider two additional mechanisms that could help explain why single women in more affluent communities tend to pay more for mortgages. The first is that women may

⁸Since student math score data is available at the school district level rather than the Census tract level, we adjust the regressions accordingly. Specifically, we replace the Census tract $\times$ origination year-month fixed effects with school district $\times$ origination year-month fixed effects. Additionally, we cluster standard errors at the school district level to account for the geographical unit at which math scores are reported.

face greater time constraints when searching for mortgage options. Mortgage search is a time-intensive process that involves learning about loan products, comparing lenders and contract terms, and carefully evaluating offers. A common hypothesis is that single women may be more time-constrained than single men due to caregiving responsibilities, such as caring for young children. However, our data do not support this explanation.

To assess this hypothesis, we merge tract-level measures of household structure from the 2017 American Community Survey (ACS) 5-year estimates into our mortgage dataset. Table 5 shows that in high-SES Census tracts, 51.2 percent of households headed by single women include children, compared with 48.7 percent for single men. In low-SES tracts, the corresponding shares are 51.6 percent and 47.1 percent. Thus, if anything, the gender gap in childcare responsibilities is slightly smaller in high-SES areas, where we observe the largest gender gaps in mortgage costs. Consistent with this pattern, specifications based on equation (3) that replace the local socioeconomic indicator with the share of single-female households with children, or with the difference between the shares of single-female and single-male households with children, yield statistically insignificant results. These findings do not support caregiving responsibilities as an explanation for the widening gender gap.

A second class of alternative explanations involves intermediary-side or institutional channels. A growing literature documents gender bias within financial institutions and differential treatment in credit markets (Egan et al., 2022; Huang et al., 2024; Delis et al., 2022), raising the possibility that some mortgage gaps could reflect lender behavior rather than borrower-side mechanisms. In the refinancing context, these channels may also operate through supply constraints that disproportionately affect marginal borrowers, among whom women may be overrepresented. During the 2020–2021 refinancing waves, intermediation markups rose, lock-to-close times lengthened, and lenders rationed or queued applications, with especially large effects for borrowers who were harder to process (Fuster et al., 2024, 2026; Frazier and Goodstein, 2025). If women’s applications in high-SES areas were systematically more difficult to process, such constraints could potentially generate larger gender gaps in refinancing.

We evaluate this explanation in several ways. First, if supply constraints were the primary mechanism, we would expect a higher concentration of marginal borrowers in low-SES areas. Consistent with this, Appendix Tables B17 and B18 show substantially higher mortgage denial rates in low-SES Census tracts than in high-SES tracts in both the 2018–2019 and 2020–2022 periods. Second, while women are more likely than men to be denied purchase mortgages, particularly in low-SES areas, denial rates for refinance loans are nearly identical across gender, which challenges a supply-side explanation for the refinancing gaps we document. Third, if lender-specific supply constraints were driving the results, controlling for lender fixed effects

should substantially attenuate the estimated gender gaps. However, as shown in Table 6, including lender fixed effects has little impact on the estimates.

Finally, in Appendix Table B19, we include lender $\times$ borrower gender $\times$ origination year fixed effects, which absorb any differential exposure of female borrowers to lender- and year-specific supply constraints. While this specification precludes estimating the main effect of gender, it still allows identification of the Female $\times$ SES interaction. The results show that larger gender gaps in high-SES areas persist even under this restrictive specification, indicating that mortgage supply constraints cannot account for the widening gender gaps we observe in more affluent communities.

4 Conclusions

This paper examines gender gaps in two of the most consequential household financial decisions: stock market participation and mortgage borrowing and refinancing. Together, these choices shape households' exposure to financial risk, the cost of housing finance, and long-run wealth accumulation. We document persistent gender differences in both domains and show that these gaps are substantially larger among higher-income and more educated individuals and communities. This pattern reveals a paradox of opportunity: expanded resources and access do not necessarily translate into more equal financial outcomes.

A central contribution of the paper is to show that this socioeconomic gradient emerges across two institutionally distinct financial decisions. Stock market participation and mortgage borrowing differ in market structure, incentives, and decision environments, yet both require households to recognize, evaluate, and act on financially valuable opportunities. The recurrence of the gradient across these settings suggests that the widening gender gaps we document are unlikely to reflect frictions specific to one market and instead point to a broader household-finance mechanism.

Our evidence points to financial literacy grounded in quantitative reasoning as an important part of this mechanism. Gender gaps in financial literacy widen with socioeconomic status and help explain why gender disparities are largest precisely in environments where participation costs are low and financial information is widely available. In the mortgage market, these gaps are concentrated in more complex concepts, such as amortization, APR, and rate-point trade-offs, and are closely related to differences in shopping behavior. The socioeconomic gradient is not explained by risk tolerance, wealth, expected future economic circumstances, or observable mortgage credit risk, and we find little support for caregiving responsibilities or the supply-side

channels we can assess as explanations for the mortgage results.

By linking adult financial behavior to childhood experiences, we show that these patterns have early roots. Gender differences in math-related attitudes and skills emerge within families, are more pronounced in high-SES households, and are associated with later differences in stock market participation. Similar patterns appear at the community level: areas with larger male-favoring gaps in student math achievement and occupational math intensity also exhibit larger gender gaps in mortgage outcomes. Together, this evidence suggests that gendered math norms and attitudes may shape financial outcomes through the accumulation and application of quantitative financial knowledge over the life cycle.

More broadly, our findings show how socioeconomic advantage can amplify, rather than mitigate, gender disparities in complex financial decisions. If the ability to benefit from financial opportunities depends partly on quantitative skills and financial reasoning shaped earlier in life, then policies focused solely on expanding access to financial markets or providing generic financial education at the point of decision may be insufficient. Our results instead point to the importance of fostering these skills before households make consequential financial decisions. Understanding these origins is important not only for explaining behavior across financial markets, but also for designing policies that improve long-run household financial well-being.

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Figures

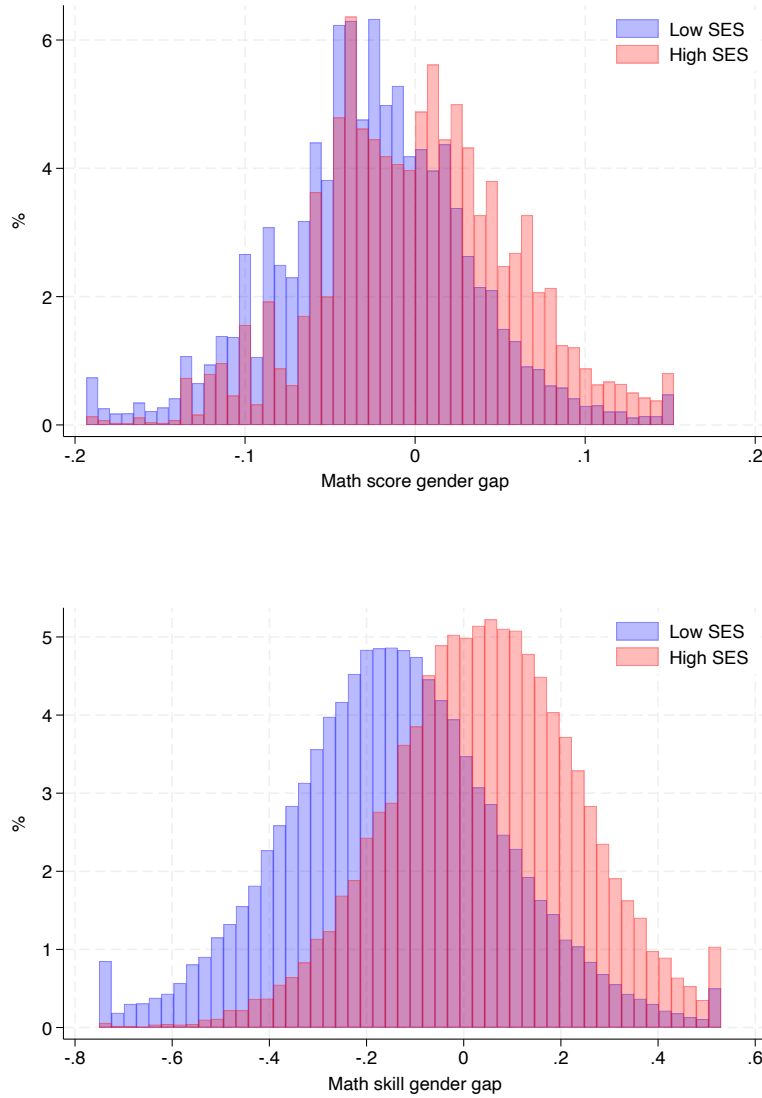


Figure 1. Gender Gap in Student Math Test Scores and Occupational Math Skill Importance by Tract SES. The top histogram shows the distribution of the student math achievement gender gap across low- and high-SES Census tracts. The bottom histogram shows the corresponding distribution for the gender gap in occupational math skill importance. Construction of the student math achievement gap and occupational math skill gap is described in Section 3.2.

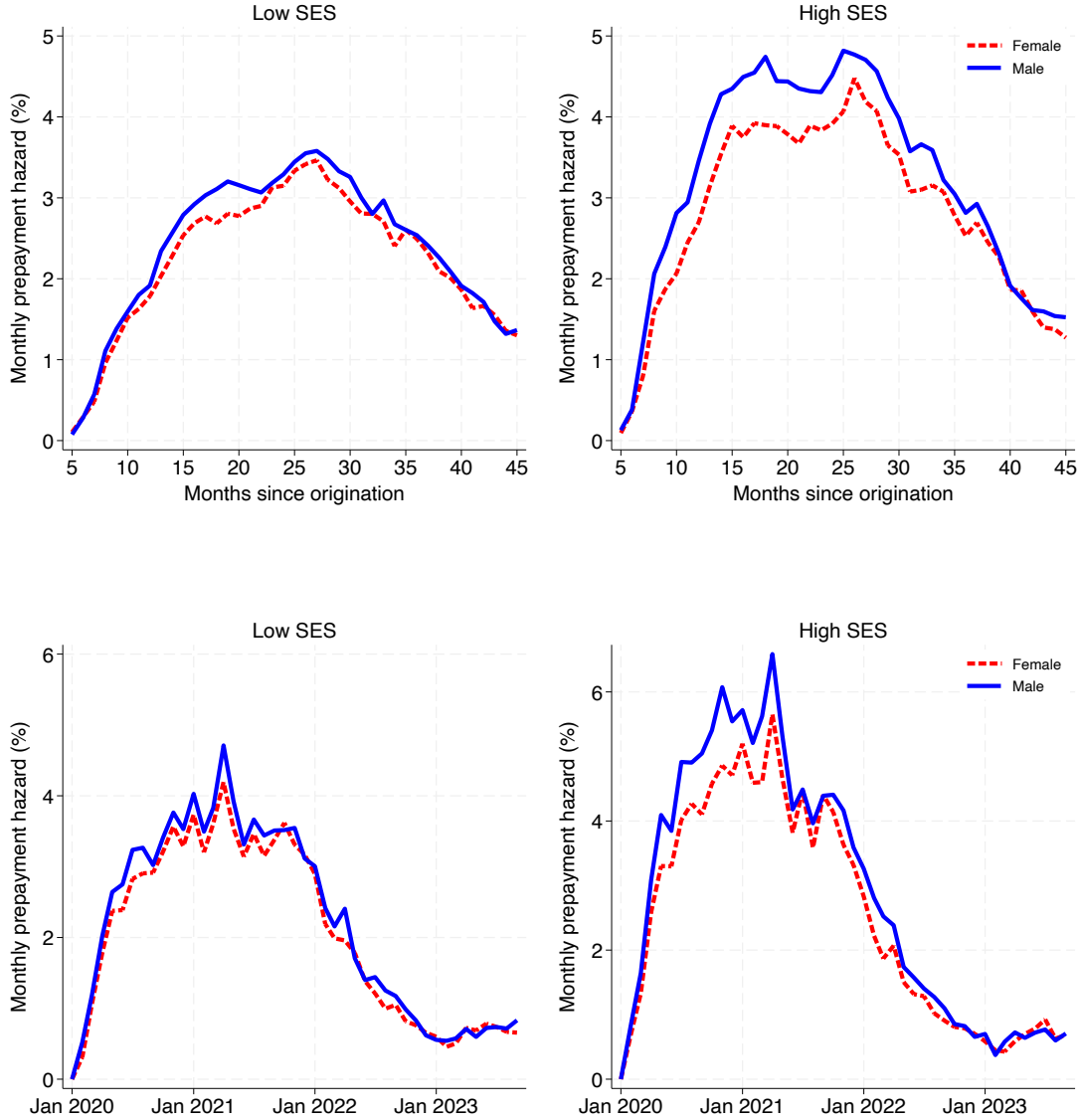


Figure 2. Kaplan–Meier Unconditional Prepayment Hazard Rates by Gender and SES. This figure plots unconditional Kaplan–Meier monthly hazard rates for mortgage prepayment by gender. The top panel presents hazards by months since origination, while the bottom panel presents hazards by calendar month. The left panels show loans in low-SES Census tracts, and the right panels show loans in high-SES Census tracts. The hazard rate is computed as the number of prepayments by gender in a given month divided by the number of outstanding mortgages for that gender at the start of the month. These estimates are unconditional on borrower and loan characteristics. The underlying data are drawn from a 15% random sample of HMDA-GSE matched loans originated between 2018 and 2019.

Tables

Table 1. Average Values of PSID Variables by SES and Gender

Family File Single Adult Panel				
	Low SES		High SES	
	Female	Male	Female	Male
Number of unique adults	6,277	5,419	3,260	3,093
Number of observations	27,379	18,431	11,366	9,045
Age (years)	46.66	39.84	44.19	40.01
Income (\$000)	22.31	31.77	63.13	77.72
Education (years)	11.90	11.79	15.19	15.14
Stock ownership (dummy)	0.05	0.05	0.17	0.22
Safe assets	50,520	53,834	139,175	198,594
Risk tolerance	0.27	0.29	0.28	0.30
Financial literacy	0.11	0.15	0.28	0.51
CDS Child-to-Adult Panel				
	Low SES		High SES	
	Female	Male	Female	Male
Number of unique children	704	750	235	234
Age in childhood (years)	11.44	11.27	11.43	11.32
Parental income (\$000)	40.95	42.44	106.89	105.90
Parental education (average years)	12.19	12.28	15.63	15.55
Math test decile	4.83	4.99	7.05	7.57
Reading test decile	5.33	4.57	7.16	6.69
Math attitude (standardized)	-0.07	0.00	-0.10	0.31
Reading attitude (standardized)	0.16	-0.19	0.23	-0.13
Number of observations (adulthood)	3,287	3,168	1,150	1,064
Age in adulthood (years)	23.87	23.99	23.79	23.75
Stock ownership (dummy)	0.04	0.07	0.17	0.23

This table reports average values of key variables used in the stock market participation analysis, separately by gender and socioeconomic status. The top panel is based on the main PSID family surveys, while the bottom panel follows individuals from the Child Development Supplement (CDS) into adulthood. Section 2.2 describes the sample construction and variable definitions.

Table 2. Gender Differences in Stock Ownership (Single Adult Panel)

	Stock Ownership									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	-0.034*** (0.005)	-0.032*** (0.004)	-0.033*** (0.004)	-0.021*** (0.004)	-0.019*** (0.003)	-0.017*** (0.003)	-0.026*** (0.006)	-0.024*** (0.006)	-0.020** (0.009)	-0.011 (0.009)
Female $\times$ Log income		-0.017*** (0.004)								
Female $\times$ Education			-0.022*** (0.005)							
Female $\times$ SES				-0.041*** (0.011)	-0.047*** (0.010)	-0.043*** (0.010)	-0.035** (0.015)	-0.036** (0.015)	-0.068*** (0.023)	-0.048** (0.023)
Safe assets						0.038*** (0.007)				
Risk tolerance								0.079*** (0.022)		
Financial literacy										0.130*** (0.017)
Sample selection	All	All	All	All	Safe assets sample		Risk tolerance sample		Fin. literacy sample	
Observations	56,395	56,395	56,395	56,395	55,246	55,246	29,358	29,358	8,786	8,786
R-squared	0.111	0.111	0.112	0.114	0.106	0.153	0.114	0.116	0.120	0.145

Standard errors, clustered at the individual level, are reported in parentheses. Log income, education, and safe asset wealth are standardized. SES is an indicator equal to one for high-SES individuals and zero otherwise. Safe asset wealth is household wealth, including housing equity, net of stock and retirement-account investments. Risk tolerance is based on the PSID Risk Tolerance Project measure (RISK96V4), which is derived from hypothetical income gambles. Financial literacy is an indicator equal to one if the respondent answers the compound-interest question (WB16K6) correctly; the financial-literacy sample includes only observations from 2016 onward. All specifications include controls for education, income, and age, as well as survey wave fixed effects. All models include the relevant linear terms in addition to the interaction terms. **p<0.05; ***p<0.01.

Table 3. Gender Differences in Stock Ownership and Math Outcomes: Sibling Comparisons (CDS Child-to-Adult Panel)

	Stock Ownership		Math Test Decile	Math Attitude	Math Test Decile
	(1)	(2)	(3)	(4)	(5)
Female	-0.032*** (0.011)	-0.033*** (0.011)	0.108 (0.195)	-0.116 (0.073)	-0.105 (0.106)
Female $\times$ SES	-0.102*** (0.036)	-0.084** (0.037)	-0.939** (0.391)	-0.331** (0.143)	
Top math		0.067** (0.031)			
Top reading		0.020 (0.032)			
Math test decile				0.111*** (0.016)	
Lagged math test decile					0.550*** (0.028)
Lagged math attitude					0.119*** (0.042)
Observations	8,638	8,638	5,616	2,604	1,599
R-squared	0.487	0.489	0.826	0.545	0.916

Standard errors, clustered at the family level, are reported in parentheses. Math attitudes are measured using the Math Ability Self-Concept variables. Math achievement is based on Broad Math Percentile Rank in 1997 and Applied Problems Percentile Rank in 2002 and 2007, as Broad Math scores are not available in later waves. Reading achievement is measured using Broad Reading Percentile Rank. All models include family fixed effects, survey wave fixed effects, and controls for age. The SES indicator is absorbed by the family fixed effects. ** $p < 0.05$; *** $p < 0.01$.

Table 4. Descriptive Statistics for HMDA-GSE Sample

	Observations	Mean	SD	Min	Max
Female	2,261,818	0.42	0.49	0.00	1.00
Mortgage rate (%)	2,261,818	4.40	0.60	2.25	7.00
Rate spread (%)	2,208,531	0.50	0.46	-0.39	2.23
Net loan cost (%)	2,261,818	1.93	1.36	-0.77	7.20
Total fees (%)	2,261,818	2.10	1.32	0.00	7.37
Total credits (%)	2,261,818	0.16	0.40	0.00	2.58
Socioeconomic status (SES)	2,261,764	0.32	0.47	0.00	1.00
Tract income (\$000)	2,261,725	72.5	28.4	5.65	250
Tract education (BA degree %)	2,261,793	35.6	17.6	0.00	96.3
SES by Reardon et al. (2019)	2,207,317	0.00	1.00	-5.97	3.09
Tract to MSA median income	2,261,818	114	38.6	35.0	257
Student math score gender gap	2,196,846	-0.01	0.05	-0.16	0.14
Occupation math skill gender gap	2,261,779	-0.05	0.22	-0.66	0.50
Credit score	2,260,505	748	46.0	309	850
Loan-to-value ratio (%)	2,261,818	76.8	17.0	2.00	389
High balance loan	2,261,818	5×10^{-5}	0.01	0.00	1.00
Investment property	2,261,818	0.03	0.18	0.00	1.00
Manufactured home	2,261,818	0.01	0.09	0.00	1.00
Subordinate loan	2,261,818	0.02	0.14	0.00	1.00
Borrower income (\$000)	2,143,893	89.5	60.6	17.0	455
Loan amount (\$000)	2,261,818	231	123	5.00	1395
Debt-to-income ratio (%)	2,256,441	36.6	9.23	1.00	53.0
Loan term (months)	2,261,818	334	61.9	60.0	360
Percentage White (%)	2,261,818	69.5	46.0	0.00	100
Share of single female HH with kids (%)	2,258,817	51.5	19.5	0.00	100
Share of single male HH with kids (%)	2,226,149	47.6	28.6	0.00	100

This table presents descriptive statistics for key variables in the mortgage analysis. The sample consists of conventional loans to single borrowers originated in HMDA in 2018–2019 and matched to Fannie Mae and Freddie Mac loan-level data. The matched data are merged with SEDA, O*NET, and ACS measures. Loan costs are expressed as a percentage of the loan amount. Rate spreads, loan costs, gender gaps, tract-to-MSA income, and borrower income are winsorized at the top and bottom 0.5%. In regressions, gender gaps, tract income, tract education, tract-to-MSA income, and SES are standardized to have mean zero and standard deviation one.

Table 5. Average Values of Mortgage Variables by SES and Gender

	Low SES tracts		High SES tracts	
Number of observations	1,527,081		734,683	
Female	0.42		0.41	
Tract income (\$000)	58.0		103	
Tract education (BA degree %)	26.8		53.9	
SES by Reardon et al. (2019)	-0.31		0.65	
Tract to MSA median income	97.7		148	
Student math score gender gap	-0.01		0.01	
Occupation math skill gender gap	-0.11		0.07	
Share of single female HH with kids (%)	51.6		51.2	
Share of single male HH with kids (%)	47.1		48.7	
	Low SES tracts		High SES tracts	
	Female	Male	Female	Male
Mortgage rate (%)	4.47	4.44	4.32	4.26
Rate spread (%)	0.57	0.56	0.37	0.34
Net loan cost (%)	2.23	2.10	1.60	1.40
Total fees (%)	2.39	2.26	1.76	1.57
Total credits (%)	0.16	0.16	0.16	0.18
Credit score	746	745	755	755
Loan-to-value ratio (%)	76.4	78.8	72.7	75.9
High balance loan	10^{-4}	10^{-4}	10^{-5}	2×10^{-5}
Investment property	0.03	0.05	0.02	0.03
Manufactured home	0.01	0.01	10^{-3}	10^{-3}
Subordinate loan	0.02	0.02	0.02	0.02
Borrower income (\$000)	70.0	85.8	96.0	122
Loan amount (\$000)	190	209	272	308
Debt-to-income ratio (%)	40.4	38.5	39.3	37.1
Loan term (months)	337	332	337	330
Percentage White (%)	69.7	69.5	72.0	67.5

This table reports average values of key variables by tract SES and borrower gender. The top panel summarizes tract-level characteristics for low- and high-SES tracts, where high-SES tracts have 2017 income and educational attainment above the sample medians. The bottom panel reports loan- and borrower-level averages separately by borrower gender within each tract SES group.

Table 6. Mortgage Cost, Borrower Gender and SES

	Mortgage Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.028*** (0.001)	0.019*** (0.001)	0.018*** (0.001)	0.042*** (0.001)	0.034*** (0.001)	0.031*** (0.001)	0.009*** (0.001)	0.005*** (0.001)	0.004*** (0.001)
Female $\times$ SES		0.026*** (0.002)	0.021*** (0.001)		0.020*** (0.001)	0.017*** (0.001)		0.010*** (0.001)	0.009*** (0.001)
R-squared	0.674	0.674	0.702	0.763	0.763	0.784	0.830	0.830	0.846
	Net Loan Cost at Origination								
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Female	0.166*** (0.002)	0.151*** (0.003)	0.143*** (0.003)	0.123*** (0.002)	0.109*** (0.003)	0.103*** (0.002)	-0.005*** (0.002)	-0.009*** (0.002)	-0.007*** (0.002)
Female $\times$ SES		0.043*** (0.004)	0.042*** (0.004)		0.036*** (0.004)	0.035*** (0.004)		0.012*** (0.004)	0.014*** (0.003)
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Lender FE	No	No	Yes	No	No	Yes	No	No	Yes
R-squared	0.453	0.453	0.533	0.510	0.510	0.583	0.622	0.622	0.695
Observations	1,826,143	1,826,135	1,825,971	1,826,143	1,826,135	1,825,971	1,706,676	1,706,668	1,706,505

Standard errors, clustered at the Census tract level, are reported in parentheses. All models include Census tract $\times$ origination year-month fixed effects. Pricing controls are based on the Fannie Mae and Freddie Mac loan-level pricing grids used to determine credit-risk adjustments to guarantee fees. They include indicators for nine loan-to-value categories, nine credit-score categories, manufactured homes, investment properties, high-balance loans, subordinate liens, loan purpose, and number of units. Additional controls include eight borrower race and ethnicity categories, income deciles, loan amount deciles, three loan-term categories, and seven borrower age groups. ***p<0.01.

Table 7. Mortgage Default, Borrower Gender and SES

	90-Day Delinquency								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	-0.451*** (0.043)	-0.432*** (0.046)	-0.412*** (0.047)	-0.402*** (0.039)	-0.379*** (0.043)	-0.368*** (0.045)	-0.446*** (0.037)	-0.458*** (0.042)	-0.453*** (0.044)
Female $\times$ SES		-0.068 (0.064)	-0.081 (0.067)		-0.069 (0.064)	-0.079 (0.068)		0.046 (0.068)	0.039 (0.072)
Origination year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Lender FE	No	No	Yes	No	No	Yes	No	No	Yes
Observations	9,986,448	9,986,283	9,986,283	9,986,448	9,986,283	9,986,283	9,980,437	9,980,272	9,980,272
R-squared	0.001	0.001	0.001	0.002	0.002	0.002	0.002	0.002	0.002

Standard errors, double-clustered at the county and origination year-month levels, are reported in parentheses. Coefficients and standard errors are multiplied by 1,000. The unit of observation is a loan-month, and the sample contains 320,939 loans. All models include origination year-month and county fixed effects. Pricing controls include indicators for nine loan-to-value categories, nine credit-score categories, manufactured homes, investment properties, high-balance loans, subordinate liens, loan purpose, and number of units. Additional controls include property value, loan age, eight borrower race and ethnicity categories, income deciles, loan amount deciles, three loan-term categories at origination, and seven borrower age groups. All models include the relevant linear terms in addition to the interaction terms. ***p<0.01.

Table 8. Mortgage Prepayment, Borrower Gender and SES

	Prepayment								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	-0.228*** (0.017)	-0.156*** (0.015)	-0.051*** (0.012)	-0.202*** (0.015)	-0.132*** (0.014)	-0.029** (0.012)	-0.042*** (0.011)	-0.003 (0.012)	0.067*** (0.014)
Female $\times$ SES		-0.210*** (0.028)	0.017 (0.031)		-0.192*** (0.027)	0.034 (0.031)		-0.126*** (0.025)	0.067* (0.035)
Female $\times$ Rate drop			-0.131*** (0.020)			-0.134*** (0.019)			-0.099*** (0.019)
Female $\times$ SES $\times$ Rate drop			-0.295*** (0.047)			-0.295*** (0.047)			-0.267*** (0.046)
Origination year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Observations	10,450,011	10,449,846	10,449,846	10,450,011	10,449,846	10,449,846	10,443,544	10,443,379	10,443,379
R-squared	0.003	0.003	0.006	0.003	0.003	0.007	0.005	0.005	0.010

Standard errors, double-clustered at the county and origination year-month levels, are reported in parentheses. Coefficients and standard errors are multiplied by 100. All models include origination year-month and county fixed effects. Pricing controls include indicators for nine loan-to-value categories, nine credit-score categories, manufactured homes, investment properties, high-balance loans, subordinate liens, loan purpose, and number of units. Additional controls include property value, loan age, eight borrower race and ethnicity categories, income deciles, loan amount deciles, three loan-term categories at origination, and seven borrower age groups. All models include the relevant linear terms and, in specifications with triple interactions, the corresponding pairwise interactions. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 9. Overall Mortgage Cost, Borrower Gender, SES and Math Skills

	Overall Cost								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.606*** (0.017)	0.807*** (0.024)	0.757*** (0.014)	0.627*** (0.016)	0.845*** (0.020)	0.752*** (0.013)	-0.025 (0.015)	0.021** (0.011)	0.045*** (0.012)
Female $\times$ SES	0.445*** (0.028)			0.367*** (0.026)			0.207*** (0.024)		
Female $\times$ Student math score gap		0.183*** (0.021)			0.148*** (0.018)			0.045*** (0.012)	
Female $\times$ Occupation math skill gap			0.177*** (0.014)			0.147*** (0.013)			0.086*** (0.012)
Location $\times$ Orig. year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Observations	1,826,135	2,159,375	1,826,137	1,826,135	2,159,375	1,826,137	1,706,668	2,044,464	1,706,670
R-squared	0.440	0.218	0.440	0.521	0.346	0.521	0.587	0.451	0.587

Standard errors are reported in parentheses and are clustered at the school-district level in columns (2), (5), and (8), and at the Census tract level otherwise. Overall cost is expressed as a percentage of the loan amount and measured over the first 10 years after origination, including origination costs, interest payments, and refinancing costs when applicable. The student math score gap is the difference between average male and female standardized math scores in grades 3–8 within each school district, using SEDA data. The occupational math skill gap is the difference between the average math skill importance of occupations held by men and women within each Census tract, constructed from O*NET math skill scores and 2017 ACS employment shares. Both math gap measures are standardized in the regressions. Columns (2), (5), and (8) include school district $\times$ origination year-month fixed effects; all other columns include Census tract $\times$ origination year-month fixed effects. Pricing controls and additional controls are as in Table 6. **p<0.05; ***p<0.01.

Table 10. Mortgage Literacy, Borrower Gender, and SES

	Knowledge Topic									
	Subprime	APR	Amortization	Points	Advanced V1	Advanced V2	Process	Fixed rate	Payment	Basic
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	-0.280*** (0.016)	-0.247*** (0.017)	-0.174*** (0.017)	-0.216*** (0.019)	-0.273*** (0.016)	-0.266*** (0.019)	-0.074*** (0.018)	-0.183*** (0.019)	-0.005 (0.018)	-0.107*** (0.018)
SES	0.083** (0.033)	0.100*** (0.034)	0.089*** (0.033)	0.103*** (0.039)	0.106*** (0.033)	0.099*** (0.038)	0.080** (0.034)	0.031 (0.035)	0.018 (0.035)	0.052 (0.034)
Female × SES	-0.123*** (0.029)	-0.143*** (0.029)	-0.173*** (0.029)	-0.154*** (0.033)	-0.172*** (0.028)	-0.184*** (0.033)	-0.150*** (0.029)	-0.020 (0.030)	-0.027 (0.030)	-0.079*** (0.029)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	29,616	29,616	29,616	21,092	29,616	21,092	29,616	29,616	29,616	29,616
R-squared	0.110	0.079	0.131	0.126	0.139	0.151	0.063	0.091	0.043	0.083

This table reports coefficient estimates from regressions of standardized mortgage-literacy measures on the female indicator, the high-SES indicator, and their interaction. Regressions use NSMO survey weights and include survey wave fixed effects, origination year-month fixed effects, and controls for income group, education, race and ethnicity, age, and credit-score category. Robust standard errors are reported in parentheses. Advanced Literacy V1 sums responses to questions on prime versus subprime loans, interest rate versus APR, and amortization. Advanced Literacy V2 also includes the question on points versus interest rates, which is available starting in wave 11. Basic Literacy sums responses to questions on the mortgage process, fixed versus adjustable rates, and the consequences of not making payments. All dependent variables are standardized to have mean zero and standard deviation one. The sample is drawn from the National Survey of Mortgage Originations and includes conventional loans to single borrowers originated between January 2013 and December 2024 across 46 survey waves. **p<0.05; ***p<0.01.

Table 11. Mortgage Shopping Behavior, Gender and Financial Literacy

	Initial Familiarity			Lender Search		
	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.149*** (0.015)		-0.010 (0.014)	-0.064*** (0.007)		-0.053*** (0.007)
Financial literacy		0.421*** (0.007)	0.420*** (0.007)		0.037*** (0.004)	0.033*** (0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	29,616	29,616	29,616	29,616	29,616	29,616
R-squared	0.143	0.278	0.278	0.037	0.039	0.041
	Loan Search			Confirm Terms		
	(7)	(8)	(9)	(10)	(11)	(12)
Female	-0.128*** (0.015)		-0.080*** (0.015)	-0.034*** (0.007)		-0.020*** (0.007)
Financial literacy		0.153*** (0.007)	0.146*** (0.008)		0.047*** (0.004)	0.046*** (0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	29,616	29,616	29,616	24,178	24,178	24,178
R-squared	0.044	0.061	0.062	0.033	0.041	0.042

This table reports coefficient estimates from regressions of mortgage search behaviors on the female indicator and financial literacy, measured using Advanced Literacy V1 from Table 10. Regressions use NSMO survey weights and include survey wave fixed effects, origination year-month fixed effects, and controls for income group, education, race and ethnicity, age, and credit-score category. Robust standard errors are reported in parentheses. The sample consists of conventional loans to single borrowers in the National Survey of Mortgage Originations, originated between January 2013 and December 2024 across 46 survey waves. The “Confirm Terms” outcome is available starting in wave 7. ***p<0.01.

INTERNET APPENDIX

APPENDIX A

A1 PSID Data and Variable Construction

A1.1 PSID Main Family Files

We use the publicly available PSID Main Family Files, obtained from the PSID Data Center at the University of Michigan.⁹ These packaged data, along with the accompanying documentation and questionnaires, provide longitudinal information on household demographics, income, and wealth, with annual observations in 1996–1997 and biennial data thereafter through 2023.

From each wave of the PSID family surveys, we extract the head’s gender, age, marital status, education (of both the head and spouse), and family income. We construct a stock market participation variable based on the question:

“Do [you/you or anyone in your family] have any shares of stock in publicly held corporations, mutual funds, or investment trusts—not including stocks in employer-based pensions or IRAs?”

We use an imputed version of the variable, which is available biennially beginning in 1999.

We merge the main family files with the Family Relationship Matrix using the family ID, interview year, and interview number, linking each interview to individual person numbers and their relationship to the head of the family. This allows us to identify the respondent. For our analysis, we retain only individuals who are heads of the family and single, ensuring that stock ownership responses reflect that individual’s behavior.

A1.2 Child Development Supplement

We use data from the PSID Child Development Supplement (CDS), obtained from the publicly available documentation and questionnaires.¹⁰ The original CDS, conducted in 1997, surveyed children up to age 12, who were subsequently re-interviewed in 2002 and 2007. The

⁹<https://simba.isr.umich.edu/Zips/zipMain.aspx>

¹⁰<https://psidonline.isr.umich.edu/CDS/Guide/documents.aspx>

CDS provides detailed measures of children’s cognitive skills and attitudes, including standardized mathematics test scores and math-related attitudes. Later CDS waves, beginning in 2014, introduced new cohorts of children; however, because these individuals have not yet reached adulthood, our analysis focuses on the original CDS sample.

From the Child Development Supplement (CDS), we extract unique identifiers for the family and the specific child surveyed, as well as the child’s gender, age, and relationship to the household head. We also extract summary measures of self-rated math and reading ability.

The math ability self-rating in 1997 and 2002 is a summary measure constructed as the mean score of responses to the following ten questions:

-
1. How good at math are you?
 2. If you were to list all the students in your class from the worst to the best in math, where would you put yourself?
 3. Some kids are better in one subject than another. For example, you might be better in math than in reading. Compared to most of your other school subjects, how good are you at math?
 4. How well do you expect to do in math this year?
 5. How good would you be at learning something new in math?
 6. How hard is math for you?
 7. Some things that you learn in school help you to do things better outside of class, that is they are useful. For example, learning about plants might help you grow a garden. In general, how useful is what you learn in math?
 8. For you, being good in math is [not important – very important].
 9. Do you find working on math assignments [very boring – very interesting (fun)]?
 10. How much do you like doing math?
-

In 2007, the math ability self-rating was constructed using four questions: questions (1), (3), (8), and (9) listed above. The reading ability self-rating measure was constructed in an analogous manner.

Children’s math achievement is measured using scores from the Woodcock–Johnson Tests of Achievement provided in the survey. In 1997, we use the Broad Math Summation Percentile Rank, constructed from the Calculation and Applied Problems subtests. In 2002 and 2007, the Calculation subtest was not administered; therefore, we use the Applied Problems Percentile Rank. Reading achievement is measured using the Broad Reading Summation Per-

centile Rank, constructed from the Letter–Word Identification and Passage Comprehension subtests.

To merge children’s data with their adult survey responses, we use family and person identifiers that uniquely identify each individual in the PSID. From the merged dataset, we impose several sample restrictions. We require that the household head is a parent of the child and that information on both parents’ education, as well as family income, is available in 1997.

To merge children’s data with their adult survey responses, we use family and person identifiers that uniquely identify each individual in the PSID dataset. From the merged dataset, we apply a small number of sample restrictions. We require that the household head is a parent of the child and that information on both parents’ education, as well as family income, is available in 1997. In our regressions, the family identifier is used to identify siblings, who share the same family identifier but have distinct person numbers. Our estimates exploit within-family variation, comparing siblings of different genders. This design effectively holds constant family background and parental resources.

A1.3 Transition into Adulthood Supplement

The analysis relies on data from the PSID Transition into Adulthood Supplement (TAS) waves, collected biannually from 2005 to 2021. These waves follow CDS respondents into adulthood and provide detailed information on education, income, and asset ownership, including stock market participation. While some of the later TAS waves added new participants, our analysis is restricted to individuals who can be tracked from childhood.

Our main variable of interest from TAS is stock market participation, measured using the survey question:

“Do you yourself have any shares of stock in publicly held corporations, mutual funds, or investment trusts, NOT including stocks in employer-based pensions or IRAs?”

This measure captures only the respondent’s personal participation in the stock market. We also use the family and person identifiers to merge TAS data with the CDS and main PSID family files.

A1.4 Wellbeing and Daily Life Supplement and Risk Tolerance Project

We use data from the 2016 PSID Wellbeing and Daily Life Supplement (WB), a self-administered module completed by household heads and spouses/partners aged 30 and older. The supplement includes a range of measures capturing psychological traits, cognitive skills, and everyday reasoning. To measure financial literacy, we focus on question WB16K6 assessing knowledge of compound interest:

“Suppose you have \$200 in a savings account. The account earns 10 percent interest each year. How much would you have in the account at the end of two years?”

We construct a dummy variable for financial literacy that takes the value 1 if the respondent answers correctly using compound interest, and 0 otherwise.

We also use a measure of risk tolerance derived from the Risk Tolerance Project embedded in the 1996 PSID wave. Following the approach of Kimball et al. (2009), we use the risk tolerance variable (RISK96V4), which is based on responses to a set of hypothetical income gamble questions administered in 1996. These questions present respondents with a series of choices between a certain income and risky alternatives that vary in potential gains and losses, allowing individuals to be ordered along a risk tolerance continuum. The Risk Tolerance Project then imputes individual estimates of risk tolerance from these responses, capturing meaningful heterogeneity in economic preferences.

TABLES FOR APPENDIX A

Table A1. Variable Definitions: PSID

Variable	Definition	Source
Stock ownership	Indicator equal to one if the respondent owns any shares of stock, mutual funds, or investment trusts held outside IRAs and other retirement accounts.	PSID Family Files
Female	Indicator equal to one if the respondent is female.	PSID Family Files
Income	Total family income in the survey year, measured in U.S. dollars.	PSID Family Files
Education	Total years of completed schooling.	PSID Family Files
SES	Indicator equal to one if the respondent has above-median income and more than 12 years of schooling.	Constructed from PSID Family Files
Financial literacy	Indicator equal to one if the respondent answers the compound interest question correctly.	PSID 2016 Wellbeing Supplement
Risk tolerance	Imputed measure of individual risk tolerance based on responses to hypothetical income gamble questions.	PSID 1996 Risk Tolerance Module
Math test decile	Childhood math achievement measured as a within-sample decile of standardized test scores.	PSID CDS
Reading test decile	Childhood reading achievement measured as a within-sample decile of standardized test scores.	PSID CDS
Math attitude	Standardized measure of confidence in, enjoyment of, and perceived ability in mathematics.	PSID CDS
Reading attitude	Standardized measure of confidence in, enjoyment of, and perceived ability in reading.	PSID CDS
High-SES family	Indicator equal to one if the respondent's family has above-median income and both parents have more than 12 years of schooling.	Constructed from PSID CDS

This table summarizes the main variables used in the PSID stock market participation analysis and the child-to-adult panel based on the Child Development Supplement (CDS). SES is defined at the individual level for the adult sample and at the family level for the child-to-adult panel.

Table A2. Future Income and Household Size (Single Adult Panel)

	Future Income		Future HH size	
	Female	Male	Female	Male
	(1)	(2)	(3)	(4)
Log income	0.498*** (0.013)	0.517*** (0.015)	0.069*** (0.018)	0.224*** (0.018)
Education	0.134*** (0.009)	0.119*** (0.011)	-0.297*** (0.024)	-0.358*** (0.027)
Age	-0.002*** (0.000)	-0.004*** (0.001)	-0.030*** (0.001)	-0.013*** (0.001)
SES	0.220*** (0.019)	0.166*** (0.023)	-0.063 (0.042)	0.053 (0.047)
Observations	23,602	14,065	24,304	14,619
R-squared	0.472	0.435	0.157	0.095

Standard errors clustered at the individual level are reported in parentheses. Future income is measured as the average of log income in the two subsequent survey waves (two and four years later). Future household size is measured similarly. All regressions include survey-wave fixed effects.

***p<0.01.

Table A3. Gender Differences in Stock Ownership (Single Adult Panel)
Controlling for Predicted Future Income and Household Size

	Stock Ownership			
	(1)	(2)	(3)	(4)
Female	-0.034*** (0.006)	-0.026*** (0.005)	-0.008 (0.015)	0.053 (0.036)
Female $\times$ SES		-0.030** (0.014)		-0.058*** (0.017)
HH size			-0.015*** (0.001)	-0.015*** (0.001)
Predicted future income			0.180*** (0.033)	0.465** (0.210)
Predicted future HH size			0.011 (0.021)	-0.001 (0.019)
Observations	33,231	33,231	33,231	33,231
R-squared	0.113	0.116	0.120	0.121

Standard errors, clustered at the individual level, are reported in parentheses. SES is an indicator equal to one for high-SES individuals and zero otherwise. Predicted future income and predicted future household size are measured using the fitted values from the first-stage regressions reported in Table A2. All specifications include individual-level controls for education, income, and age, as well as survey wave fixed effects. In addition to the interaction terms, all models include the linear terms. **p<0.05; ***p<0.01.

Table A4. Average Values of PSID Variables by SES and Gender
Excluding Lowest 25th Income and Education Percentiles

Family File Single Adult Panel				
	Low SES		High SES	
	Female	Male	Female	Male
Number of unique adults	4,077	3,453	3,260	3,093
Number of observations	12,455	9,197	11,366	9,045
Age (years)	45.99	39.21	44.19	40.01
Income (\$000)	31.99	43.01	63.13	77.72
Education (years)	12.89	12.62	15.19	15.14
Stock ownership (dummy)	0.08	0.07	0.17	0.22
Safe assets	73,800	70,757	139,174	198,594
Risk tolerance	0.26	0.29	0.28	0.30
Financial literacy	0.15	0.22	0.28	0.51

This table replicates the top panel of Table 1, excluding individuals in the bottom 25th percentile of the income and education distributions every year. The original SES cutoffs for the low- and high-SES groups are maintained.

Table A5. Gender Differences in Stock Ownership (Single Adult Panel)
Excluding Lowest 25th Income and Education Percentiles

	Stock Ownership									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	-0.034*** (0.006)	-0.018*** (0.006)	-0.021*** (0.005)	-0.011* (0.006)	-0.011* (0.005)	-0.011** (0.005)	-0.014* (0.008)	-0.012 (0.008)	-0.005 (0.014)	0.004 (0.014)
Female × Log income		-0.031*** (0.010)								
Female × Education			-0.031*** (0.010)							
Female × SES				-0.042*** (0.011)	-0.046*** (0.011)	-0.042*** (0.011)	-0.035** (0.015)	-0.036** (0.015)	-0.066*** (0.025)	-0.051** (0.024)
Safe assets						0.030*** (0.006)				
Risk tolerance								0.089*** (0.028)		
Financial literacy										0.116*** (0.018)
Sample selection	All	All	All	All	Safe assets sample		Risk tolerance sample		Fin. literacy sample	
Observations	36,716	36,716	36,716	36,716	35,754	35,754 21,702	21,702	6,073	6,073	
R-squared	0.124	0.125	0.125	0.127	0.121	0.133	0.123	0.124	0.133	0.151

Standard errors, clustered at the individual level, are reported in parentheses. Log income, education and safe asset wealth are standardized. SES is an indicator equal to one for high-SES individuals and zero otherwise. Risk tolerance estimates in PSID (RISK96V4) are derived from hypothetical income gambles. Financial literacy is an indicator equal to one if the respondent answers compound interest question WB16K6 correctly and zero otherwise. The financial literacy sample only includes observations post-2016. Safe asset wealth is the household's wealth including housing equity minus the value of stock and retirement account investments. All specifications include individual-level controls for education, income, and age, as well as survey wave fixed effects. In addition to the interaction terms, all models include the linear terms. *p<0.10; **p<0.05; ***p<0.01.

Table A6. Gender Differences in Stock Ownership (Single Adult Panel)
Excluding Lowest 25th Income and Education Percentiles
Logit and Probit Models

	Stock Ownership	
	(1)	(2)
	Logit Model	Probit Model
Female	-0.131 (0.089)	-0.091** (0.045)
Female $\times$ SES	-0.229** (0.109)	-0.125** (0.056)
Observations	36,716	36,716
R-squared	0.161	0.162

Standard errors, clustered at the individual level, are reported in parentheses. SES is an indicator equal to one for high-SES individuals and zero otherwise. All specifications include individual-level controls for education, income, and age, as well as survey wave fixed effects. In addition to the interaction terms, all models include the linear terms. **p<0.05.

APPENDIX B

B1 HMDA Data

The Home Mortgage Disclosure Act (HMDA) requires financial institutions to publicly disclose loan-level information about mortgage applications and originations. The disclosures contain demographic information, such as the applicant’s gender and race, and essential details such as the applicant’s income, the loan amount, lender name, and the Census tract of the property. Overall, this dataset includes 90% of mortgage originations in the US and has been used extensively in previous studies. The HMDA data serves multiple purposes, one of which is to provide information for evaluating whether financial institutions are effectively addressing the housing needs of all members within their communities.

In response to the 2010 Dodd-Frank Wall Street Reform and Consumer Protection Act (Dodd-Frank Act) and the 2015 definitive rule from the Consumer Financial Protection Bureau (CFPB), the HMDA data was augmented to incorporate additional details on mortgage loans with the purpose to enhance understanding of the mortgage market. The implementation and commencement of data collection under the revised HMDA data framework began on January 1, 2018.

This expanded dataset provides comprehensive information on loan pricing, covering aspects such as interest rates, discount points, lender credits, points and fees, and negative amortization. Furthermore, it incorporates factors related to applicant creditworthiness, including debt-to-income ratio and loan-to-value (LTV) ratio, as well as specifics about the financed property, such as whether it involves manufactured housing with or without land. These newly added factors are commonly used by lenders in their loan decision-making processes and are essential controls for our study.

Utilizing the expanded HMDA dataset, our sample includes mortgages originated on or after January 1, 2018. We conclude our sample period for HMDA loans on December 31, 2019, avoiding the highly volatile and extraordinary years that ensued due to COVID-19 policies affecting the mortgage market. Our final HMDA dataset consists of all conventional first-lien mortgage originations for single-family properties between January 1, 2018, and December 31, 2019.

The expanded HMDA data provides deeper insights into how borrowers and lenders interact in the mortgage market by offering more detailed information on mortgage pricing. However, a key limitation is the absence of borrower credit scores in the publicly available

data. To address this, we merge HMDA loans with loan-level datasets from Fannie Mae and Freddie Mac (the GSEs), which include borrower credit scores. The merged dataset not only incorporates credit scores but also provides precise mortgage origination month and year, more granular debt-to-income ratios, and, crucially for refinancing analysis, the monthly performance of loans. These additional variables from the GSE data significantly enhance the depth and accuracy of our analysis. Since there is no unique identifier to connect the loans in the different datasets, we develop a very thorough algorithm to merge them.

B2 HMDA-GSE Matching Procedure

Below we describe the steps to match the mortgages from Fannie Mae and Freddie Mac, the Government Sponsored Enterprises (GSEs), to the HMDA database.

B2.1 HMDA Data Processing

We begin with all loans in the HMDA 2018 and 2019 files. We keep loans secured by a first lien, for single family homes (1 to 4 units), either site-built or manufactured. We remove reverse mortgages, open-end lines of credit, loans with negative amortization, interest-only, loans with balloon payments and other non-amortizing features. We remove the loans for which the Census tract, the interest rate, the total loan cost or the loan term is missing. Finally, we keep the loans that were originated (action taken = 1) or were purchased (action taken = 6). The total number of mortgages in this raw database is 15,592,356, with 3,272,471 being purchased loans.

B2.1.1 Matching purchased loans with originated loans

Purchased loans are loans that were purchased by a financial institution after originated by another financial institution. These include closed mortgage loans where the purchasing entity has not reviewed the application or made a credit decision prior to purchasing. With the expansion of the HMDA reporting requirements, entities purchasing covered loans are required to collect, record and report on the new and modified data points for purchases that occur on or after January 1, 2018.

However, the entities covered by the extended requirements for loan purchases are not required to report certain data for purchased loans. For some types of data, such as the

rate spread, a covered purchaser complies with the HMDA requirements by reporting “not applicable.” For certain other types of data, such as the borrower’s income, age, ethnicity, race and sex, a covered purchaser may, but is not required to, report such data, and can either report “not applicable” or provide the data if it so elects.

To link purchased loans to GSE loans where applicable, we first need to address the issue of missing key loan characteristics in the purchased loan data. Many critical fields required for both matching and future analysis, such as borrower gender, credit score, and rate spread, are often absent. Therefore, we must match purchased loans with their corresponding originated loans in the HMDA dataset to recover these missing data fields. These fields are essential for accurately conducting the HMDA-GSE matching process and for supporting our subsequent analysis.

We match the originated loans with the purchased loans on the following ten variables, imposing an exact match for all variables: Census tract, loan amount, interest rate, total loan cost, loan term, conforming loan limit, construction method, occupancy type, number of units and loan type. This results in 2,170,157 matched loans from the 3,272,471 purchased loans (66.3% match rate).

Not all purchased loans have a match, as some of them were probably originated in 2017, and sold in 2018, and we were only matching the loans originated in 2018 or 2019. Moreover, we impose very strict matching conditions. A slight difference in the reported loan amount or interest rate would prevent a match even if it were the same loan.

We do the following post-match tests to ensure correct matches. We drop matched loans if one of the following applies:

- (a) The origination year is after the year of purchase
- (b) The absolute difference between the property value of the originated and purchased loan is larger than \$5,000
- (c) The absolute difference between the borrower income of the originated and purchased loan is larger than \$5,000
- (d) One of the loans in a matched pair is reported as a high-cost mortgage and the other as not a high-cost mortgage
- (e) The matched loans do not have the same loan purpose (home purchase, home improvement, refinancing or cash-out refinancing) when the loan purpose is different from “Other purpose” or “Not applicable.”

The next step ensures that we only keep unique matches. Originated loans that match to multiple purchased loans might have been sold multiple times. If an originated loan matches to multiple purchased loans we keep the purchased loan that is sold to Fannie Mae or Freddie Mac, or the purchaser type is “Not applicable”. After this cleaning procedure, 99.91% of the loans have a unique match. We drop the remaining 0.09% originated or purchased loans that match to more than one loan. In total, 1,922,181 originated loans uniquely match to purchased loans in HMDA between 2018 and 2019.

B2.1.2 Updating the HMDA dataset

Once we have the unique matches of the purchased loans to the originated loans, we replace the demographic variables of the purchased loans, which were mostly missing, with the demographic variables of the matched originated loans. These variables include the applicant and co-applicant ethnicity, race, sex, age, credit score type, and various loan characteristics, such as combined LTV, rate spread, total points and fees, origination charges, discount points, lender credits, prepayment penalty term, intro rate period, debt to income ratio, direct submission of application indicator, initially payable to institution indicator, automated underwriting system used and indicator for preapproval request.

The updated HMDA dataset contains the originated mortgages that were not sold to another financial institution before they were potentially sold to the GSEs. In addition, for those mortgages that were sold to another financial institution, the updated dataset replaces the originated loans with the matched purchased loans, since these will be matched in the next step to the loans in the GSE databases. The total number of unique mortgages in the updated HMDA dataset is 13,670,175.

B2.2 GSE Data Processing

We put together the mortgages acquired by Fannie Mae or Freddie Mac, from January 1st, 2018 to December 31st, 2020, from quarterly data files, publicly available on the websites of the GSEs. The mortgage database contains mortgage characteristics such as the month and year of origination, the interest rate, the 3-digit zip code and the identifier of the financial institution that originated the mortgage. In this initial database there are 8,195,899 mortgages from Fannie Mae and 6,988,080 from Freddie Mac.

B2.3 HMDA-GSE Matching

We create two separate files from the updated HMDA data. The first one, to be matched with loans from Fannie Mae, keeps the mortgages for which the purchaser type is Fannie Mae. The second one, to be matched with loans from Freddie Mac, keeps the mortgages for which the purchaser type is Freddie Mac. We also do a second round matching with the purchaser type being “Not applicable”, which improves the percentage of GSE loans matched. Throughout the full process we impose strict matching criteria over multiple loan characteristics, and ensure that the matches are unique.

B2.3.1 Lenders

We match the loans on several dimensions. First, we match the financial institution in HMDA (legal entity identifier) that holds the loan to the financial institution in the GSE dataset (seller) that sold the loan. The matching of the financial institutions uses the HMDA lender file for the years 2018 and 2019. This crosswalk file, called “The Avery file”, was constructed by Robert Avery and contains matching information for all lenders who have ever filed a HMDA report. The HMDA ID is matched to information in the FRB’s NIC system and FFIEC or TFR Call Reports for each filing year.

We manually check the matching of lenders and adjust for mergers and acquisitions of lenders within the time frame of the loan originations. Since the seller identifiers in the GSE data are set to “Other sellers” for smaller financial institutions, we allow the matching algorithm to match “Other sellers” to a number of legal entity identifiers in HMDA.

B2.3.2 Geographical units

Second, we match the Census tract identifier in HMDA, with the 3-digit zip code in the GSE datasets. To do this, we utilize the 2018 crosswalk file from the U.S. Department of Housing and Urban Development (HUD). At this stage we keep all the observations, even if a Census tract belongs to more than one 3-digit zip code. Most of the duplicated HMDA loans that were created in the process of matching them to multiple zip codes are eliminated at later stages since we require matching on many other loan characteristics.

B2.3.3 Other loan characteristics

We make the following adjustments to the HMDA data, to improve potential matches with the GSE loans:

- (a) We create a new combined LTV variable, to follow the GSE reporting rule: The result of the calculation of the LTV must be truncated (shortened) to two decimal places, then rounded up to the nearest whole percent. For example, 94.01% is converted to 95%, and 80.001% is converted to 80%.
- (b) We create a new variable for the number of borrowers, that takes the values of 1, 2 or missing, based on the values of the applicant sex and co-applicant sex.

We match the mortgages in the updated HMDA dataset to the mortgages in the Fannie Mae and Freddie Mac datasets on the following variables, in addition to the lender and geographical unit we described earlier, imposing an exact match for these five variables: interest rate, loan term at origination, combined LTV, number of borrowers and occupancy type (principal residence, second residence or investment property).

Next, we perform post-match tests and we require that the loans match on a number of additional characteristics. For these characteristics we allow some margins for the values in the HMDA and GSE databases not to be exactly equal.

- (a) We drop loans for which the origination year in HMDA is after the GSE acquisition year.
- (b) We drop loans for which the absolute difference between the loan amount in HMDA and the loan amount in the GSE database is larger than \$5,000
- (c) We require the following matching of the loan purpose from the two databases: Loans with purpose in HMDA of 1 (home purchase) should match with the purpose from GSEs of “P” (purchase), loans with purpose in HMDA of 31 (refinancing) should match with the purpose from GSEs of “R” (refinance), and loans with purpose in HMDA of 32 (cash-out refinancing) should match with the purpose from GSEs of “C” (cash-out refinance). Other values of loan purpose from HMDA (2 - home improvement, 4 - other purpose or 5 - not applicable) are let to match with any of the three purposes from GSEs.
- (d) We require the following matching of the origination channel used by the party that delivered the loan to the issuer. In the GSE databases the origination channel is either correspondent (a loan that is originated by a party other than a mortgage loan seller and

is then sold to a mortgage loan seller), retail (a loan for which the mortgage loan seller takes the loan application and then processes, underwrites, funds, and sells the loan to the GSE) or broker. We drop loans if the channel is correspondent and the action taken in HMDA is 1 (loan originated). Also, we drop loans if the channel is retail and the action taken in HMDA is 6 (purchased loan).

- (e) We require the following matching of the debt-to-income ratio (DTI) in HMDA that is reported either as an integer or as a bucket of values to the DTI in the GSE databases that is reported as an integer. The GSE reporting rule for DTI is as follows: The DTI value is rounded down to the nearest whole number if the decimal is less than 0.5, and rounded up if the decimal is 0.5 or greater.

DTI from GSEs (%)	DTI in HMDA
<20	"<20%"
20	"<20%" or "20%-<30%"
>20 and <30	"20%-<30%"
30	"20%-<30%" or "30%-<36%"
>30 and <36	"30%-<36%"
36	"30%-<36%" or "36" or "37"
37	"36" or "37" or "38"
38	"37" or "38" or "39"
...	...
48	"47" or "48" or "49"
49	"48" or "49" or "50%-60%"
50	"49" or "50%-60%"
>50 and <60	"50%-60%"
60 or 61	"50%-60%" or ">60%"
>61	">60%"

- (f) We require matching of the number of total units, 1 to 4, in the two databases.
- (g) We keep only conventional loans and drop FHA, FSA/RHS and VA loans. After the matching and the previous additional checks more than 99.99% of the matched loans are conventional, which is a confirmation that the algorithm matches the HMDA loans to the loans acquired by Fannie and Freddie correctly.

After we implement the above matching algorithm, we require that each loan from HMDA is matched to a unique loan from either Fannie or Freddie and vice versa. Out of the matched

loans from Fannie 96.95% are unique, and out of the matched loans from Freddie 97.76% are unique. Out of the HMDA loans matched to Fannie 95.95% are unique, and out of the HMDA loans matched to Freddie 97.24% are unique. We remove from the database the remaining loans that have multiple matches. The HMDA-GSE matched dataset contains 4,197,266 loans.

B3 Occupation Math Skill Gap: Data and Variable Construction

The occupation math skill gap measures the relative importance of math skills in the occupations held by men versus women within each Census tract. We develop this novel variable by combining the occupational composition of males and females in each Census tract with the math skill importance assigned to each occupation, using it as an indicator of gender disparities in the adult population’s ability to apply mathematical skills to problem-solving. These skills are crucial for effective mortgage decisions, such as securing lower costs, evaluating trade-offs, or refinancing at optimal times. In this section, we describe the datasets and calculations used to construct the occupation math skill gap measure.

The two primary data sources for constructing this variable are the Occupational Information Network (O*NET), which provides detailed scores on the math skills and knowledge required for specific occupations, and the American Community Survey (ACS), which offers Census tract employment data by gender. Because O*NET scores are available for a highly granular set of occupations, while ACS data are at broader occupation levels, we use various crosswalks to connect these datasets.

O*NET is a comprehensive database that provides detailed information about U.S. occupations, relying on large-scale surveys of workers, employers, and experts. Specifically, we use the May 2017 database from the O*NET Database Releases Archive. Using the 2017 occupation classification ensures the maximum match with the 2017 occupation codes from the Bureau of Labor Statistics (BLS).

The O*NET database contains multiple files that quantify different characteristics and abilities required for each occupation. We use the files for “Skills” and “Knowledge”, and focus on the importance and level of mathematics in each one. We describe the construction of the measure for math skill importance, and replicate the steps and analysis for math skill level, math knowledge level and math knowledge importance.

The O*NET “Skills” file contains a list of 964 occupations (8-digit occupation codes) and reports a “math skill importance” score for each one. This score ranges from 1 to 5, and measures the degree to which mathematical problem solving abilities are required to perform the tasks associated with the specific occupation. For example, Mathematicians score 5/5 in math skill importance, Financial Analysts 3.9/5, Sales Managers 3/5 and Actors 1/5.

Using the occupation codes, we merge the O*NET dataset with the number of people employed in each occupation from the 2017 Occupational Employment and Wage Statistics (OEWS) Tables from BLS. The BLS dataset classifies occupations in various levels of detail, specifically in major, minor, broad and detailed groups. We keep the occupations in the most refined classification, which is the detailed groups. This results in 809 6-digit occupation codes.

We merge the occupation codes from O*NET with the occupation codes from BLS. Before merging, we process the 8-digit codes in O*NET to create 6-digit codes as follows: If the O*NET code ends in “00” we keep that occupation and remove any other occupation with the same 6-digit code (and different last two digits). We do this because the codes ending in digits other than “00” are subsets of the main occupation, e.g. 11-1011.00 is the code for Chief Executives and 11-1011.03 is the code for Chief Sustainability Officers. For occupations with the same 6-digit codes that do not have a representative occupation ending in “00” we allocate the average value of the math skill importance to the 6-digit code. After creating the 6-digit codes, we do a visual check of the occupation descriptions in both databases, and do manual adjustments to the codes, so that the same occupation description always corresponds to the same 6-digit occupation code in both databases. The merge of O*NET math skill importance dataset with the BLS employment data results in 765 unique 6-digit occupation codes.

Next, we obtain table S2401 with Census tract employment data by gender from the 2017 ACS 5-year estimates. This dataset contains descriptions of 25 occupations, which we match with major occupation groups in BLS that correspond to 2- or 3-digit codes. We merge the ACS data with the O*NET-BLS dataset, using the 25 major occupation group codes.

To calculate the gender gap in math skill importance in each Census tract we follow these steps: First, we calculate the weighted average of the math skill importance for the major occupation groups, using as weights the 2017 national employment figures from BLS. Second, we calculate the weighted average math skill importance for males and females in each Census tract, using as weights the employment numbers for each gender in the ACS. Finally, we calculate the gap as the math skill importance score of males minus females.

FIGURES FOR APPENDIX B

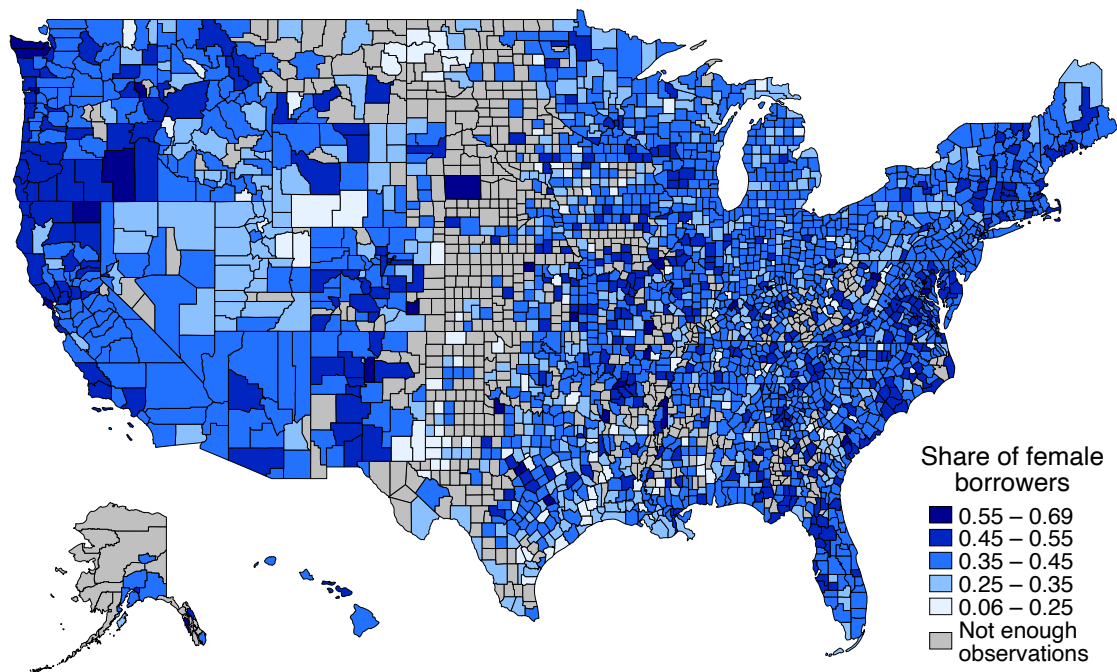


Figure B1. Geographical Distribution of Single Female versus Single Male Borrowers. The map illustrates the share of single female borrowers among all single borrowers across U.S. counties, derived from the merged HMDA-GSE loan dataset. The share is calculated for counties with at least 20 loan observations in the sample.

TABLES FOR APPENDIX B

Table B1. Variable Definitions: HMDA-GSE Merged Sample

Variable	Definition	Source
Mortgage rate	Contract interest rate on the mortgage at origination.	HMDA
Rate spread	Difference between the loan's annual percentage rate (APR) and the average prime offer rate for a comparable transaction on the date the rate is set.	HMDA
Total fees	Total points and fees, divided by the loan amount at origination.	HMDA
Total credits	Total lender credits, divided by the loan amount at origination.	HMDA
Net loan cost	Total points and fees minus lender credits, divided by the loan amount at origination.	HMDA
Prepayment	Indicator equal to one if the loan is voluntarily paid off in full before maturity.	Fannie Mae, Freddie Mac
Default	Indicator equal to one if the loan becomes 90 days or more past due.	Fannie Mae, Freddie Mac
Overall mortgage cost	Hypothetical 10-year mortgage cost, including origination costs and interest payments, as well as refinancing costs if the loan prepays within the 10-year horizon.	Constructed from HMDA-GSE
Female	Indicator equal to one if the borrower is female.	HMDA
Tract income	Median household income in the Census tract.	ACS
Tract education	Share of adults aged 25 and older in the Census tract with a bachelor's degree or higher.	ACS
SES	Indicator equal to one if tract median income and tract educational attainment both exceed the sample median.	Constructed from HMDA-GSE and ACS
Rate drop	Indicator equal to one if the prevailing market mortgage rate in a given month is below the market rate in the origination month.	Freddie Mac PMMS
Student math score gender gap	Difference between average male and female standardized math scores in a given school district.	SEDA
Occupation math skill gender gap	Difference between male and female average math-skill importance in a given tract, based on O*NET occupations.	O*NET and ACS

This table summarizes the variables used in the mortgage pricing, refinancing, and default analyses based on the merged HMDA–GSE sample. HMDA loan-level data are merged with Fannie Mae and Freddie Mac loan performance data and supplemented with Census tract characteristics from the American Community Survey (ACS), student achievement data from the Stanford Education Data Archive (SEDA), and occupational skill measures from O*NET.

Table B2. Average Values of Mortgage Variables by Gender and Tract Income

	Low income tracts		High income tracts	
Female	0.43		0.40	
Tract income (\$000)	50.7		94.3	
Tract education (BA degree %)	25.4		45.8	
Socioeconomic status	-0.49		0.48	
Tract to MSA median income	91.1		137	
Student math score gender gap	-0.02		0.002	
Occupation math skill gender gap	-0.12		0.02	
Share of single female HH with kids (%)	52.6		50.4	
Share of single male HH with kids (%)	47.0		48.2	

	Low income tracts		High income tracts	
	Female	Male	Female	Male
Mortgage rate (%)	4.49	4.47	4.35	4.30
Rate spread (%)	0.60	0.60	0.41	0.39
Net loan cost (%)	2.35	2.23	1.70	1.52
Total fees (%)	2.51	2.39	1.85	1.68
Total credits (%)	0.16	0.16	0.16	0.17
Credit score	745	744	752	752
Loan-to-value ratio (%)	77.0	79.2	73.4	76.5
High balance loan	10^{-4}	10^{-4}	2×10^{-5}	2×10^{-5}
Investment property	0.03	0.05	0.02	0.03
Manufactured home	0.01	0.01	0.003	0.003
Subordinate loan	0.02	0.02	0.02	0.02
Borrower income (\$000)	66.9	82.3	90.2	112
Loan amount (\$000)	176	192	258	290
Debt-to-income ratio (%)	37.1	35.8	37.6	36.2
Loan term (months)	336	332	337	331
Percentage White (%)	70.1	69.8	70.8	67.9

This table reports average values of the key variables in the study for low- and high-income tracts, and by gender. The top panel shows the percentage of female borrowers and average values at the tract level. Low-income tracts are defined as those with tract income in 2017 below the median, while high-income tracts are defined as those with tract income above the median. The bottom panel shows average values at the loan and borrower level for male and female borrowers in low- and high-income tracts. The variables are as described in Table 4. $N = 2,261,818$.

Table B3. Average Values of Mortgage Variables by Gender and Tract Education

	Low education tracts		High education tracts	
Female	0.41		0.42	
Tract income (\$000)	56.3		88.7	
Tract education (BA degree %)	21.3		50.0	
Socioeconomic status	-0.39		0.39	
Tract to MSA median income	93.5		135	
Student math score gender gap	-0.02		0.002	
Occupation math skill gender gap	-0.14		0.04	
Share of single female HH with kids (%)	52.0		51.0	
Share of single male HH with kids (%)	47.5		47.7	
	Low education tracts		High education tracts	
	Female	Male	Female	Male
Mortgage rate (%)	4.49	4.47	4.35	4.29
Rate spread (%)	0.60	0.60	0.41	0.38
Net loan cost (%)	2.34	2.20	1.74	1.53
Total fees (%)	2.50	2.36	1.89	1.70
Total credits (%)	0.16	0.16	0.16	0.17
Credit score	743	743	754	754
Loan-to-value ratio (%)	76.6	79.0	73.9	76.6
High balance loan	6×10^{-5}	8×10^{-5}	3×10^{-5}	2×10^{-5}
Investment property	0.03	0.05	0.02	0.03
Manufactured home	0.01	0.02	0.002	0.002
Subordinate loan	0.02	0.02	0.02	0.02
Borrower income (\$000)	66.2	80.8	89.9	115
Loan amount (\$000)	182	200	249	285
Debt-to-income ratio (%)	37.5	36.3	37.1	35.8
Loan term (months)	336	332	337	331
Percentage White (%)	67.9	68.2	72.9	69.5

This table reports average values of the key variables in the study for low- and high-education tracts, and by gender. The top panel shows the percentage of female borrowers and average values at the tract level. Low-education tracts are defined as those where the share of the tract population over 25 years old with a bachelor's degree or higher is below the median, while high-education tracts are defined as those where this share is above the median. The bottom panel shows average values at the loan and borrower level for male and female borrowers in low- and high-education tracts. The variables are as described in Table 4. N = 2,261,818.

Table B4. Pairwise Correlations of Local Socioeconomic Indicators

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Socioeconomic status (SES)						
(2) Tract median income	0.738					
(3) Tract education (BA degree %)	0.719	0.716				
(4) SES by Reardon et al. (2019)	0.450	0.576	0.440			
(5) Tract to MSA median income	0.614	0.756	0.687	0.336		
(6) Student math score gender gap	0.205	0.242	0.193	0.517	0.180	
(7) Occupation math skill gender gap	0.392	0.386	0.489	0.294	0.328	0.195

This table presents pairwise correlations of the socioeconomic variables and gender gaps in math test scores and math skills that we use in the study. $N = 2,194,370$.

Table B5. Dispersion in Mortgage Costs by SES

	Mean	SD	P25	P75	IQR
Low SES tracts					
Mortgage rate (%)	4.45	0.60	3.99	4.88	0.89
Rate spread (%)	0.57	0.47	0.24	0.82	0.58
Net loan cost (%)	2.15	1.42	1.18	2.91	1.73
Total fees (%)	2.31	1.38	1.33	3.04	1.71
Total credits (%)	0.16	0.40	0.00	0.10	0.10
High SES tracts					
Mortgage rate (%)	4.28	0.58	3.88	4.75	0.88
Rate spread (%)	0.35	0.40	0.08	0.55	0.47
Net loan cost (%)	1.48	1.10	0.77	2.02	1.26
Total fees (%)	1.65	1.05	0.93	2.16	1.23
Total credits (%)	0.17	0.39	0.00	0.13	0.13

This table reports summary statistics for the main mortgage cost variables separately for low- and high-SES Census tracts. For each variable, we report the mean, standard deviation (SD), 25th percentile (P25), 75th percentile (P75), and interquartile range (IQR), which is the difference between the 75th and 25th percentiles. High SES tracts are defined as those with tract income and tract education in 2017 above the median, while the rest are low SES tracts. The sample consists of conventional mortgages with a single borrower originated in 2018–2019 from the merged HMDA–GSE dataset. Results are qualitatively and quantitatively similar when dispersion is first computed within Census tract and then averaged across tracts within each SES group.

Table B6. Descriptive Statistics for Single and Joint Mortgages by Gender

	Single borrower		Primary borrower in joint mortgage	
	Female	Male	Female	Male
Percentage of borrowers (%)	23.2	32.2	11.7	32.9
Mortgage rate (%)	4.42	4.38	4.37	4.33
Rate spread (%)	0.51	0.49	0.47	0.42
Net loan cost (%)	2.03	1.86	1.68	1.62
Total fees (%)	2.19	2.03	1.83	1.77
Total credits (%)	0.16	0.16	0.15	0.15
Credit score	749	748	748	753
Loan-to-value ratio (%)	75.3	77.8	76.4	74.5
High balance loan	4×10^{-5}	5×10^{-5}	3×10^{-5}	2×10^{-5}
Investment property	0.03	0.04	0.02	0.03
Manufactured home	0.01	0.01	0.01	0.01
Subordinate loan	0.02	0.02	0.02	0.02
Borrower income (\$000)	78.3	98.0	125	128
Loan amount (\$000)	216	242	267	271
Debt-to-income ratio (%)	40.0	38.0	36.2	36.2
Loan term (months)	337	332	330	326
Percentage White (%)	70.4	68.8	74.0	76.6

This table reports average values of the key variables in the study for all loans in the HMDA-GSE matched data, by borrower gender. For mortgages with two borrowers, averages are shown separately for male and female primary borrowers, as HMDA records gender and race for both applicants and co-applicants. Loan cost variables are expressed as a percentage of the loan amount. Rate spread, loan costs, and borrower income are winsorized at the top and bottom 0.5% to minimize outliers. Source: HMDA data for conventional loans originated in 2018–2019, matched with loans from Fannie Mae and Freddie Mac. N = 4,092,384.

Table B7. Descriptive Statistics for Single and Joint Mortgages by SES and Gender

	Single borrower		Primary borrower in joint mortgage		Single borrower		Primary borrower in joint mortgage	
	Female	Male	Female	Male	Female	Male	Female	Male
	Low SES tracts				High SES tracts			
Percentage of borrowers (%)	24.5	33.3	11.3	30.9	21.2	30.7	12.3	35.8
Mortgage rate (%)	4.47	4.45	4.43	4.40	4.33	4.27	4.28	4.25
Rate spread (%)	0.58	0.57	0.56	0.50	0.39	0.35	0.36	0.32
Net loan cost (%)	2.27	2.13	1.91	1.86	1.64	1.43	1.38	1.32
Total fees (%)	2.43	2.29	2.06	2.00	1.79	1.61	1.53	1.48
Total credits (%)	0.15	0.16	0.15	0.14	0.16	0.18	0.16	0.15
Credit score	745	745	744	750	754	754	753	758
Loan-to-value ratio (%)	76.6	78.9	77.6	75.7	73.1	76.1	74.8	72.9
High balance loan	1×10^{-4}	1×10^{-4}	4×10^{-5}	3×10^{-5}	1×10^{-5}	2×10^{-5}	1×10^{-5}	3×10^{-6}
Investment property	0.03	0.05	0.03	0.04	0.02	0.03	0.02	0.02
Manufactured home	0.01	0.01	0.01	0.01	2×10^{-3}	1×10^{-3}	1×10^{-3}	1×10^{-3}
Subordinate loan	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.03
Borrower income (\$000)	68.9	84.5	111	113	94.1	119	143	146
Loan amount (\$000)	186	205	231	233	266	301	315	319
Debt-to-income ratio (%)	40.4	38.5	36.8	36.8	39.3	37.2	35.5	35.3
Loan term (months)	337	332	330	326	337	330	331	326
Percentage White (%)	0.69	0.69	0.73	0.76	0.72	0.68	0.75	0.77

This table reports average values of the key variables in the study for all loans in the HMDA-GSE matched data for low- and high-SES tracts and by gender. Low- and high-SES tracts are constructed as in Table 5. The variables are as described in Table B6. N = 4,092,384.

Table B8. Mortgage Cost, Borrower Gender and SES: **Alternative Outcomes**

	Rate Spread								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.017*** (0.001)	0.008*** (0.001)	0.006*** (0.001)	0.024*** (0.001)	0.018*** (0.001)	0.015*** (0.001)	0.007*** (0.001)	0.003*** (0.001)	0.001 (0.001)
Female $\times$ SES		0.024*** (0.002)	0.021*** (0.002)		0.018*** (0.001)	0.015*** (0.001)		0.012*** (0.001)	0.011*** (0.001)
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Lender FE	No	No	Yes	No	No	Yes	No	No	Yes
Observations	1,773,253	1,773,245	1,773,088	1,773,253	1,773,245	1,773,088	1,656,279	1,656,271	1,656,114
R-squared	0.404	0.404	0.470	0.615	0.615	0.660	0.660	0.660	0.704
	Total Fees			Total Credits					
	(10)	(11)	(12)	(13)	(14)	(15)			
Female	-0.009*** (0.002)	-0.012*** (0.002)	-0.007*** (0.002)		-0.004*** (0.001)	-0.002** (0.001)	0.000 (0.001)		
Female $\times$ SES		0.007* (0.004)	0.010*** (0.003)			-0.005*** (0.002)	-0.005*** (0.001)		
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes		Yes	Yes	Yes		
Pricing controls	Yes	Yes	Yes		Yes	Yes	Yes		
Additional controls	Yes	Yes	Yes		Yes	Yes	Yes		
Lender FE	No	No	Yes		No	No	Yes		
Observations	1,706,676	1,706,668	1,706,505		1,706,676	1,706,668	1,706,505		
R-squared	0.638	0.638	0.705		0.328	0.328	0.419		

Standard errors, clustered at the Census tract level, are in parentheses. Pricing controls, additional controls and fixed effects are as in Table 6. *p<0.10; **p<0.05; ***p<0.01.

Table B9. Mortgage Cost, Borrower Gender and Local Factors: **Alternative SES Measures**

	Mortgage Rate			Net Loan Cost at Origination		
	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.027*** (0.001)	0.041*** (0.001)	0.008*** (0.001)	0.164*** (0.002)	0.121*** (0.002)	-0.006*** (0.002)
Female $\times$ Tract income	0.013*** (0.001)	0.008*** (0.001)	0.005*** (0.001)	0.021*** (0.002)	0.013*** (0.002)	0.008*** (0.002)
Female	0.027*** (0.001)	0.040*** (0.001)	0.008*** (0.001)	0.164*** (0.002)	0.120*** (0.002)	-0.006*** (0.002)
Female $\times$ Tract education	0.014*** (0.001)	0.011*** (0.001)	0.005*** (0.001)	0.023*** (0.002)	0.023*** (0.002)	0.009*** (0.002)
Female	0.027*** (0.001)	0.041*** (0.001)	0.008*** (0.001)	0.164*** (0.002)	0.120*** (0.002)	-0.006*** (0.002)
Female $\times$ Tract/MSA income	0.014*** (0.001)	0.011*** (0.001)	0.005*** (0.001)	0.028*** (0.002)	0.025*** (0.002)	0.003* (0.002)
R-squared	0.674	0.763	0.830	0.453	0.510	0.622
Observations	1,826,143	1,826,143	1,706,676	1,826,143	1,826,143	1,706,676
Female	0.028*** (0.001)	0.043*** (0.001)	0.007*** (0.001)	0.179*** (0.004)	0.147*** (0.003)	-0.010*** (0.002)
Female $\times$ SES by Reardon	0.015*** (0.001)	0.010*** (0.001)	0.003*** (0.001)	0.042*** (0.004)	0.029*** (0.003)	0.010*** (0.002)
R-squared	0.546	0.677	0.768	0.219	0.299	0.496
Observations	2,165,397	2,165,397	2,050,122	2,165,397	2,165,397	2,050,122
Pricing controls	No	Yes	Yes	No	Yes	Yes
Additional controls	No	No	Yes	No	No	Yes

Standard errors, clustered at the Census tract level (or at the school district level for regressions using Reardon et al.'s SES), are in parentheses. All specifications include Census tract $\times$ origination year and month fixed effects; in the Reardon SES regressions, we instead include school district $\times$ origination year and month fixed effects. Tract income is measured as median Census tract income in 2017, and tract education is measured as the share of the tract population aged 25 and older with a bachelor's degree or higher in 2017, both from ACS. SES (Reardon et al., 2019) is the composite socioeconomic status measure from the SEDA dataset, constructed at the school district level. Tract-MSA income is defined as the ratio of Census tract median income to MSA median income, using HMDA data. All SES measures are standardized to have mean zero and standard deviation one. Pricing controls and additional controls are as in Table 6. * $p < 0.10$; *** $p < 0.01$.

Table B10. Borrower Gender, SES and Loan Cost: **White Borrowers**

	Mortgage Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.024*** (0.001)	0.019*** (0.001)	0.018*** (0.001)	0.041*** (0.001)	0.037*** (0.001)	0.034*** (0.001)	0.008*** (0.001)	0.006*** (0.001)	0.005*** (0.001)
Female $\times$ SES		0.014*** (0.002)	0.012*** (0.002)		0.011*** (0.002)	0.009*** (0.002)		0.006*** (0.001)	0.006*** (0.001)
R-squared	0.696	0.696	0.721	0.779	0.779	0.798	0.840	0.840	0.855
	Net Loan Cost at Origination								
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Female	0.170*** (0.003)	0.162*** (0.004)	0.160*** (0.003)	0.125*** (0.003)	0.119*** (0.003)	0.116*** (0.003)	-0.013*** (0.002)	-0.013*** (0.003)	-0.008*** (0.003)
Female $\times$ SES		0.021*** (0.006)	0.024*** (0.005)		0.018*** (0.005)	0.021*** (0.005)		0.000 (0.005)	0.005 (0.004)
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Lender FE	No	No	Yes	No	No	Yes	No	No	Yes
R-squared	0.465	0.465	0.545	0.521	0.521	0.594	0.637	0.637	0.709
Observations	1,166,496	1,166,494	1,166,323	1,166,496	1,166,494	1,166,323	1,085,565	1,085,563	1,085,383

Standard errors, clustered at the Census tract level, are in parentheses. Pricing controls, additional controls and fixed effects are as in Table 6. The sample contains only White borrowers. ***p<0.01.

Table B11. Borrower Gender, SES and Loan Cost: **By Age Group**

	Mortgage rate		Rate spread		Net loan cost	
	(1)	(2)	(3)	(4)	(5)	(6)
Age ≤ 34 (N=302,410)						
Female	0.019*** (0.002)	0.004** (0.002)	0.004** (0.002)	-0.002 (0.002)	0.056*** (0.006)	-0.012** (0.006)
Female $\times$ SES	0.018*** (0.003)	0.012*** (0.003)	0.019*** (0.003)	0.012*** (0.003)	0.039*** (0.010)	0.020** (0.009)
R-squared	0.814	0.874	0.694	0.760	0.559	0.701
Age 35-44 (N=268,901)						
Female	0.030*** (0.002)	0.010*** (0.002)	0.017*** (0.002)	0.008*** (0.002)	0.076*** (0.008)	-0.009 (0.007)
Female $\times$ SES	0.030*** (0.004)	0.012*** (0.003)	0.023*** (0.003)	0.013*** (0.003)	0.046*** (0.011)	0.023** (0.009)
R-squared	0.812	0.877	0.687	0.752	0.573	0.715
Age 45-54 (N=202,086)						
Female	0.030*** (0.003)	0.001 (0.003)	0.011*** (0.003)	-0.001 (0.003)	0.092*** (0.009)	-0.015** (0.008)
Female $\times$ SES	0.020*** (0.004)	0.008** (0.004)	0.013*** (0.004)	0.010*** (0.004)	0.041*** (0.013)	0.021** (0.010)
R-squared	0.799	0.874	0.677	0.752	0.586	0.742
Age 55-64 (N=123,630)						
Female	0.028*** (0.004)	-0.002 (0.003)	0.012*** (0.003)	-0.001 (0.003)	0.112*** (0.012)	-0.002 (0.009)
Female $\times$ SES	0.017*** (0.006)	0.003 (0.005)	0.011** (0.006)	0.002 (0.005)	0.089*** (0.018)	0.026* (0.015)
R-squared	0.803	0.878	0.676	0.759	0.598	0.769
Age ≥ 65 (N=57,640)						
Female	0.038*** (0.005)	0.003 (0.004)	0.011** (0.005)	0.001 (0.004)	0.115*** (0.017)	-0.004 (0.014)
Female $\times$ SES	0.011 (0.009)	0.003 (0.008)	0.012 (0.009)	0.003 (0.008)	0.090*** (0.030)	0.048** (0.024)
R-squared	0.810	0.876	0.651	0.758	0.598	0.781
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	Yes	No	Yes	No	Yes
Lender FE	No	Yes	No	Yes	No	Yes

Standard errors, clustered at the Census tract level, are in parentheses. Pricing controls, additional controls and fixed effects are as in Table 6. *p<0.10; **p<0.05; ***p<0.01.

Table B12. Mortgage Cost, Borrower Gender and SES: **Purchase Loans**

	Mortgage Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.018*** (0.001)	0.008*** (0.001)	0.005*** (0.001)	0.034*** (0.001)	0.026*** (0.001)	0.022*** (0.001)	0.007*** (0.001)	0.003*** (0.001)	0.001 (0.001)
Female $\times$ SES		0.026*** (0.002)	0.022*** (0.002)		0.022*** (0.002)	0.019*** (0.002)		0.012*** (0.002)	0.011*** (0.001)
R-squared	0.700	0.700	0.733	0.781	0.781	0.807	0.832	0.832	0.854
	Net Loan Cost at Origination								
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Female	0.131*** (0.003)	0.115*** (0.004)	0.109*** (0.004)	0.107*** (0.003)	0.095*** (0.004)	0.086*** (0.003)	-0.001 (0.003)	-0.002 (0.003)	-0.006** (0.003)
Female $\times$ SES		0.043*** (0.006)	0.039*** (0.006)		0.034*** (0.006)	0.032*** (0.005)		0.004 (0.005)	0.006 (0.005)
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Lender FE	No	No	Yes	No	No	Yes	No	No	Yes
R-squared	0.485	0.485	0.559	0.526	0.526	0.599	0.630	0.630	0.700
Observations	927,934	927,930	927,750	927,934	927,930	927,750	858,039	858,035	857,855

Standard errors, clustered at the Census tract level, are in parentheses. Pricing controls, additional controls and fixed effects are as in Table 6. The sample contains only purchase loans (59.5% of all loans). **p<0.05; ***p<0.01.

Table B13. Mortgage Cost, Borrower Gender and SES: **Refinance Loans**

	Mortgage Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.047*** (0.002)	0.037*** (0.002)	0.032*** (0.002)	0.054*** (0.001)	0.047*** (0.002)	0.042*** (0.002)	0.011*** (0.001)	0.008*** (0.001)	0.005*** (0.001)
Female $\times$ SES		0.024*** (0.003)	0.018*** (0.003)		0.018*** (0.003)	0.013*** (0.003)		0.009*** (0.002)	0.007*** (0.002)
R-squared	0.682	0.682	0.710	0.773	0.773	0.792	0.848	0.848	0.864
	Net Loan Cost at Origination								
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Female	0.195*** (0.004)	0.175*** (0.005)	0.153*** (0.005)	0.134*** (0.004)	0.117*** (0.005)	0.104*** (0.005)	-0.009** (0.004)	-0.018*** (0.005)	-0.013*** (0.004)
Female $\times$ SES		0.049*** (0.008)	0.049*** (0.007)		0.042*** (0.008)	0.043*** (0.007)		0.022*** (0.007)	0.025*** (0.006)
Tract $\times$ Orig. yr-m FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Lender FE	No	No	Yes	No	No	Yes	No	No	Yes
R-squared	0.547	0.547	0.635	0.598	0.598	0.675	0.690	0.690	0.763
Observations	554,451	554,451	554,221	554,451	554,451	554,221	517,047	517,047	516,811

Standard errors, clustered at the Census tract level, are in parentheses. Pricing controls, additional controls and fixed effects are as in Table 6. The sample contains only refinance loans (40.5% of all loans). **p<0.05; ***p<0.01.

Table B14. Mortgage Prepayment, Borrower Gender and Local Factors: **Alternative SES Measures**

	Prepayment					
	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.211*** (0.015)	-0.051*** (0.012)	-0.179*** (0.014)	-0.020* (0.011)	-0.044*** (0.011)	0.074*** (0.014)
Female $\times$ Tract income	-0.098*** (0.012)	0.010 (0.011)	-0.088*** (0.012)	0.020* (0.011)	-0.066*** (0.012)	0.026* (0.014)
Female $\times$ Tract income $\times$ Rate drop		-0.141*** (0.018)		-0.142*** (0.018)		-0.127*** (0.017)
Female	-0.230*** (0.016)	-0.052*** (0.013)	-0.199*** (0.014)	-0.008 (0.012)	-0.047*** (0.011)	0.098*** (0.015)
Female $\times$ Tract education	-0.095*** (0.014)	-0.008 (0.015)	-0.088*** (0.014)	0.017 (0.012)	-0.056*** (0.013)	0.034** (0.014)
Female $\times$ Tract education $\times$ Rate drop		-0.128*** (0.021)		-0.139*** (0.019)		-0.126*** (0.018)
Female	-0.225*** (0.017)	-0.049*** (0.012)	-0.196*** (0.015)	-0.022* (0.012)	-0.043*** (0.012)	0.088*** (0.015)
Female $\times$ SES by Reardon	-0.098*** (0.013)	0.016 (0.010)	-0.086*** (0.012)	0.027*** (0.009)	-0.056*** (0.010)	0.041*** (0.011)
Female $\times$ SES by Reardon $\times$ Rate drop		-0.150*** (0.020)		-0.151*** (0.020)		-0.135*** (0.019)
Female	-0.218*** (0.016)	-0.049*** (0.012)	-0.186*** (0.015)	-0.019 (0.012)	-0.045*** (0.011)	0.081*** (0.015)
Female $\times$ Tract/MSA income	-0.079*** (0.012)	0.018 (0.012)	-0.071*** (0.012)	0.027** (0.012)	-0.035*** (0.012)	0.048*** (0.015)
Female $\times$ Tract/MSA income $\times$ Rate drop		-0.131*** (0.020)		-0.132*** (0.020)		-0.118*** (0.019)
Pricing controls	No	No	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	Yes	Yes
Observations	10,450,011	10,450,011	10,450,011	10,450,011	10,443,544	10,443,544
R-squared	0.003	0.006	0.003	0.007	0.005	0.010

Standard errors, clustered at the county and origination year-month levels, are in parentheses. Coefficients and standard errors are multiplied by 100. All models include origination year-month and county fixed effects. Controls and fixed effects are as in Table 8. In addition to the interaction terms, all models include the linear terms and double interaction terms. *p<0.10; **p<0.05; ***p<0.01.

Table B15. Overall Mortgage Cost, Borrower Gender and Local Factors: **Alternative SES Measures**

	Overall Cost								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.769*** (0.014)	0.744*** (0.014)	0.744*** (0.014)	0.761*** (0.013)	0.744*** (0.013)	0.740*** (0.013)	0.051*** (0.012)	0.039*** (0.012)	0.039*** (0.012)
Female × Tract income		0.239*** (0.014)			0.169*** (0.013)			0.118*** (0.012)	
Female × Tract education			0.234*** (0.014)			0.208*** (0.013)			0.116*** (0.012)
Observations	1,826,143	1,826,129	1,826,137	1,826,143	1,826,129	1,826,137	1,706,676	1,706,662	1,706,670
R-squared	0.440	0.440	0.440	0.521	0.521	0.522	0.587	0.587	0.587
Female		0.805*** (0.022)	0.748*** (0.014)		0.842*** (0.019)	0.743*** (0.013)		0.020** (0.010)	0.043*** (0.012)
Female × SES by Reardon		0.296*** (0.020)			0.209*** (0.017)			0.097*** (0.012)	
Female × Tract/MSA income			0.240*** (0.014)			0.202*** (0.013)			0.089*** (0.012)
Observations		2,165,397	1,826,143		2,165,397	1,826,143		2,050,122	1,706,676
R-squared		0.219	0.440		0.346	0.522		0.452	0.587
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes

Standard errors, clustered at the Census tract level or at the school district level for the SES by Reardon regressions, are in parentheses. All specifications include Census tract × origination year and month fixed effects; in the Reardon SES regressions, we instead include school district × origination year and month fixed effects. Overall cost is expressed as a percentage of the loan amount and is calculated for 10 years after origination. The overall cost includes origination cost, interest, and additional cost if the loan gets refinanced within this interval. Pricing controls, additional controls and fixed effects are as in Table 6. **p<0.05; ***p<0.01.

Table B16. Overall Mortgage Cost, Borrower Gender, Math and Other Skills

	Overall Cost				
	(1)	(2)	(3)	(4)	(5)
Female	0.844*** (0.020)	0.739*** (0.013)	0.739*** (0.013)	0.739*** (0.013)	0.739*** (0.013)
Female $\times$ Student math score gap	0.126*** (0.021)				
Female $\times$ Student reading score gap	0.044** (0.022)				
Female $\times$ Occupation math skill gap		0.067*** (0.018)	0.074*** (0.018)	0.052*** (0.020)	0.060*** (0.018)
Female $\times$ Occupation listening skill gap		-0.010 (0.021)			
Female $\times$ Occupation speaking skill gap			-0.024 (0.019)		
Female $\times$ Occupation reading skill gap				0.016 (0.020)	
Female $\times$ Occupation writing skill gap					0.005 (0.018)
Female $\times$ Tract education		0.182*** (0.017)	0.186*** (0.016)	0.175*** (0.015)	0.177*** (0.015)
Observations	2,159,303	1,826,137	1,826,137	1,826,137	1,826,137
R-squared	0.346	0.522	0.522	0.522	0.522

Standard errors, in parentheses, are clustered at the school district level in Model (1) and at the Census tract level in all other models. Overall cost is expressed as a percentage of the loan amount and is calculated over the first 10 years after origination. Student math and reading score gaps are defined as the difference between average male and female standardized test scores in grades 3–8 within each school district, using SEDA data. Occupation math skill gaps measure the difference between the average importance of math skills in male- versus female-held occupations within each Census tract, where skill importance from O*NET is weighted by 2017 gender-specific employment shares across major occupation groups. Gaps in active listening, speaking, reading comprehension, and writing skills are constructed analogously. All score and skill gap measures are standardized to have mean zero and unit standard deviation. All models include Census tract $\times$ origination year–month fixed effects, except Model (1), which includes school district $\times$ origination year–month fixed effects, and all models include Pricing controls as in Table 6. **p<0.05; ***p<0.01.

Table B17. Mortgage Application Outcomes 2018 - 2019

			Percentage of originated applications (%)					
			Closed	Withdrawn	Denied	Approved	Preapproval	
			because	by	by	but not	request	approved
			Number	incomplete	applicant	lender	denied	but not
			originated			accepted		accepted
All Loans								
All	Female	1,894,578	6.52	23.38	21.63	4.21	0.07	0.11
Tracts	Male	2,796,214	7.33	24.57	20.39	4.35	0.06	0.10
Low	Female	1,145,181	7.13	23.69	25.90	4.64	0.09	0.12
SES	Male	1,620,910	7.89	24.61	24.21	4.74	0.08	0.10
High	Female	749,319	5.59	22.91	15.11	3.57	0.03	0.09
SES	Male	1,175,205	6.56	24.51	15.11	3.82	0.03	0.09
Purchase Loans								
All	Female	1,135,740	4.11	18.32	14.85	3.98	0.11	0.18
Tracts	Male	1,706,435	4.41	18.74	13.31	3.99	0.09	0.16
Low	Female	697,402	4.91	18.25	18.74	4.46	0.15	0.20
SES	Male	1,006,006	5.12	18.06	16.45	4.38	0.13	0.17
High	Female	438,275	2.85	18.42	8.66	3.22	0.05	0.15
SES	Male	700,358	3.38	19.72	8.78	3.42	0.05	0.15
Refinance Loans								
All	Female	758,838	10.12	30.96	31.78	4.56	0.00	0.00
Tracts	Male	1,089,779	11.92	33.70	31.47	4.93	0.00	0.00
Low	Female	447,779	10.58	32.16	37.06	4.91	0.00	0.00
SES	Male	614,904	12.43	35.34	36.91	5.31	0.00	0.00
High	Female	311,044	9.45	29.24	24.18	4.06	0.00	0.00
SES	Male	474,847	11.26	31.58	24.44	4.43	0.00	0.00

This table presents the number of applications that resulted in mortgage originations and the different outcomes of mortgage applications, as a percentage of originated mortgages, by applicant gender and by Census tract SES. The sample contains conventional mortgage applications by single applicants in 2018 and 2019 from HMDA.

Table B18. Mortgage Application Outcomes 2020 - 2022

			Percentage of originated applications (%)					
			Closed	Withdrawn	Denied	Approved	Preapproval	Preapproval
Number			because	by	by	but not	request	approved
originated			incomplete	applicant	lender	accepted	denied	but not
								accepted
All Loans								
All	Female	4,927,715	8.47	23.68	18.81	4.06	0.04	0.16
Tracts	Male	7,352,022	9.34	25.29	17.34	4.24	0.03	0.16
Low	Female	2,740,420	9.23	24.04	22.32	4.43	0.06	0.18
SES	Male	3,862,134	9.97	25.30	20.50	4.57	0.04	0.17
High	Female	2,187,120	7.52	23.23	14.41	3.59	0.03	0.13
SES	Male	3,489,624	8.65	25.28	13.84	3.88	0.02	0.14
Purchase Loans								
All	Female	2,018,612	5.34	21.18	16.84	4.12	0.11	0.39
Tracts	Male	2,988,424	5.03	22.18	13.57	3.99	0.09	0.39
Low	Female	1,165,001	6.56	21.08	21.03	4.68	0.13	0.42
SES	Male	1,644,381	5.85	21.20	16.58	4.45	0.11	0.41
High	Female	853,486	3.67	21.32	11.11	3.34	0.07	0.34
SES	Male	1,343,858	4.01	23.39	9.89	3.44	0.06	0.36
Refinance Loans								
All	Female	2,909,103	10.64	25.42	20.18	4.02	0.00	0.00
Tracts	Male	4,363,598	12.30	27.42	19.92	4.41	0.00	0.00
Low	Female	1,575,419	11.20	26.23	23.28	4.24	0.00	0.00
SES	Male	2,217,753	13.02	28.34	23.40	4.65	0.00	0.00
High	Female	1,333,634	9.98	24.45	16.51	3.76	0.00	0.00
SES	Male	2,145,766	11.55	26.47	16.32	4.16	0.00	0.00

This table presents the number of applications that resulted in mortgage originations and the different outcomes of mortgage applications, as a percentage of originated mortgages, by applicant gender and by Census tract SES. The sample contains conventional mortgage applications by single applicants from 2020 to 2022 from HMDA.

Table B19. Overall Mortgage Cost, Borrower Gender, SES and Math Skills: **Lender-Gender-Year FE**

	Overall Cost								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female $\times$ SES	0.354*** (0.027)			0.295*** (0.026)			0.158*** (0.024)		
Female $\times$ Student math score gap		0.184*** (0.019)			0.144*** (0.017)			0.049*** (0.011)	
Female $\times$ Occupation math skill gap			0.148*** (0.014)			0.123*** (0.013)			0.068*** (0.012)
Location $\times$ Orig. year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lender $\times$ Gender $\times$ Orig. year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Observations	1,825,402	2,158,593	1,825,404	1,825,402	2,158,593	1,825,404	1,705,892	2,043,675	1,705,894
R-squared	0.473	0.269	0.473	0.546	0.379	0.546	0.607	0.478	0.607

Standard errors in parentheses, clustered at the schools district level in models (2), (5) and (8) and at the Census tract level in the rest of the models. All variable are as in Table 9. Models (2), (5) and (8) include school district $\times$ origination year-month fixed effects. The rest of the models include Census tract $\times$ origination year-month fixed effects. Additionally, all models include lender $\times$ borrower gender $\times$ origination year fixed effects. Pricing controls and additional controls are as in Table 6. ***p<0.01.

APPENDIX C

C1 PISA Data and Variable Construction

We use data from the U.S. version of the 2012 Program for International Student Assessment (PISA) Student Questionnaire, administered by the OECD. PISA surveys 15-year-old students and collects detailed information on academic performance, socioeconomic background, beliefs and attitudes toward learning. Following Breda et al. (2020), we construct a standardized measure of math beliefs and norms that combines students' perceived control over their success in mathematics with their parents' beliefs about the importance of mathematics for future careers. These measures are constructed conditional on students' math ability.

While Breda et al. (2020) develop country-level measures of gender stereotypes, we adapt their approach to the individual level using U.S. student responses. This allows us to analyze how gender differences in math norms vary with parental education and household wealth. We also use information from the student questionnaire on gender, math ability, wealth, and parents' educational attainment.

Our measure is based on responses to the following two questions in the PISA 2012 Student Questionnaire:

1. "Thinking about your mathematics lessons: To what extent do you agree with the following statement? Whether or not I do well in mathematics is completely up to me."
2. "Thinking about how people important to you view mathematics: How strongly do you agree with the following statements? My parents believe that mathematics is important for my career."

Responses range from 1 (Strongly agree) to 4 (Strongly disagree). We reverse this scale so that higher values indicate stronger agreement with perceived control over mathematics performance and stronger parental beliefs in the importance of mathematics for students' careers. Next, we standardize these values so that their weighted mean (using PISA student weights to ensure U.S. sample representativeness) is zero, and their weighted standard deviation is one. To construct a combined measure, we take the average of the two standardized values and standardize it again, accounting for survey weights.

The parents' education variables capture the highest level of formal education attained

by the student’s mother and father, categorized according to the International Standard Classification of Education (ISCED):

- We group ISCED levels 0 (None), 1 (Primary education), and 2 (Lower secondary education) into the category “High school or lower.” This category predominantly consists of individuals with lower secondary education, as fewer than 4% of the sample attained only primary education, and fewer than 2% had no formal education. ISCED level 3 (Upper secondary education) had zero observations in our sample.
- We group ISCED levels 4 (Post-secondary non-tertiary education) and 5 (Vocational higher education) into “Some college.”
- ISCED level 6 (Bachelor’s degree or equivalent) is the maximum level in the sample and forms the category “College graduate.”

These categories align with those used in our financial literacy analysis with the NSMO data.

The index of family wealth measures students’ access to material resources at home, based on responses about household assets. Specifically, it considers whether students have a room of their own, a link to the internet, a dishwasher, a DVD player, a guest room, a high speed internet connection, and a musical instrument. Additionally, it includes responses on the number of cellular phones, televisions, computers, cars, and rooms with a bath or shower. As with previous variables, we standardize the family wealth index so that its weighted mean is zero and its weighted standard deviation is one.

C2 Social Norms and Math Gender Stereotypes

To examine how gender differences in math norms vary with parental education and household wealth, we estimate the following regression specification:

$$N_i = \beta_1 Female_i + \sum_k \beta_{2,k} ParentEdu_{i,k} + \sum_k \beta_{3,k} Female_i \times ParentEdu_{i,k} + \beta_4 MathAchievement_i + \epsilon_i, \quad (C1)$$

where N_i represents the standardized average of the responses to the two math beliefs and norms questions for student i . $ParentEdu_{i,k}$ represents indicators for the educational attainment of the student’s mother and father. In a third regression we replace parental education with a standardized index of family wealth.

PISA does not report actual math scores for each student but instead provides five plausible values, which are randomly drawn from the distribution of possible scores based on the student’s responses. Any estimation involving students’ math ability (in our case, the regression model in equation (C1)) requires computing the target statistic for each plausible value, applying the appropriate student weights, and averaging the five resulting estimates. Standard errors are computed using a replication method that accounts for PISA’s stratified two-stage sampling design (selection of schools and students within schools). PISA provides 80 alternative sets of individual weights and detailed guidelines for their use. Corrected standard errors are obtained via bootstrap techniques, requiring the regression of interest to be run separately for each of the five plausible values—first using the primary set of individual weights, then using the 80 alternative weight sets. The final point estimate is the average of the five regressions using the primary weights, while the standard error incorporates both measurement and sampling errors. This is implemented in Stata using the “repest” command, provided by OECD.

Figure C1 presents the estimated levels of math beliefs and norms across different levels of parental education and wealth and Table C1 presents the results. We find that among students from lower-education and lower-wealth backgrounds, girls exhibit stronger math beliefs than boys, though the gender differences are not statistically significant. However, as parental education and wealth increase, boys’ math norms strengthen, while girls’ math norms remain stable or decline. This divergence leads to a statistically and economically significant gender gap of approximately 0.2 standard deviations among students from wealthier families and those with college-educated parents.¹¹

¹¹Breda et al. (2020) note that in the United States, gender essentialist norms are stronger in high-income households than in low-income households, though they do not further document or elaborate on this finding. Parental income is not available in the publicly accessible version of our dataset, preventing us from directly examining this relationship.

FIGURES FOR APPENDIX C

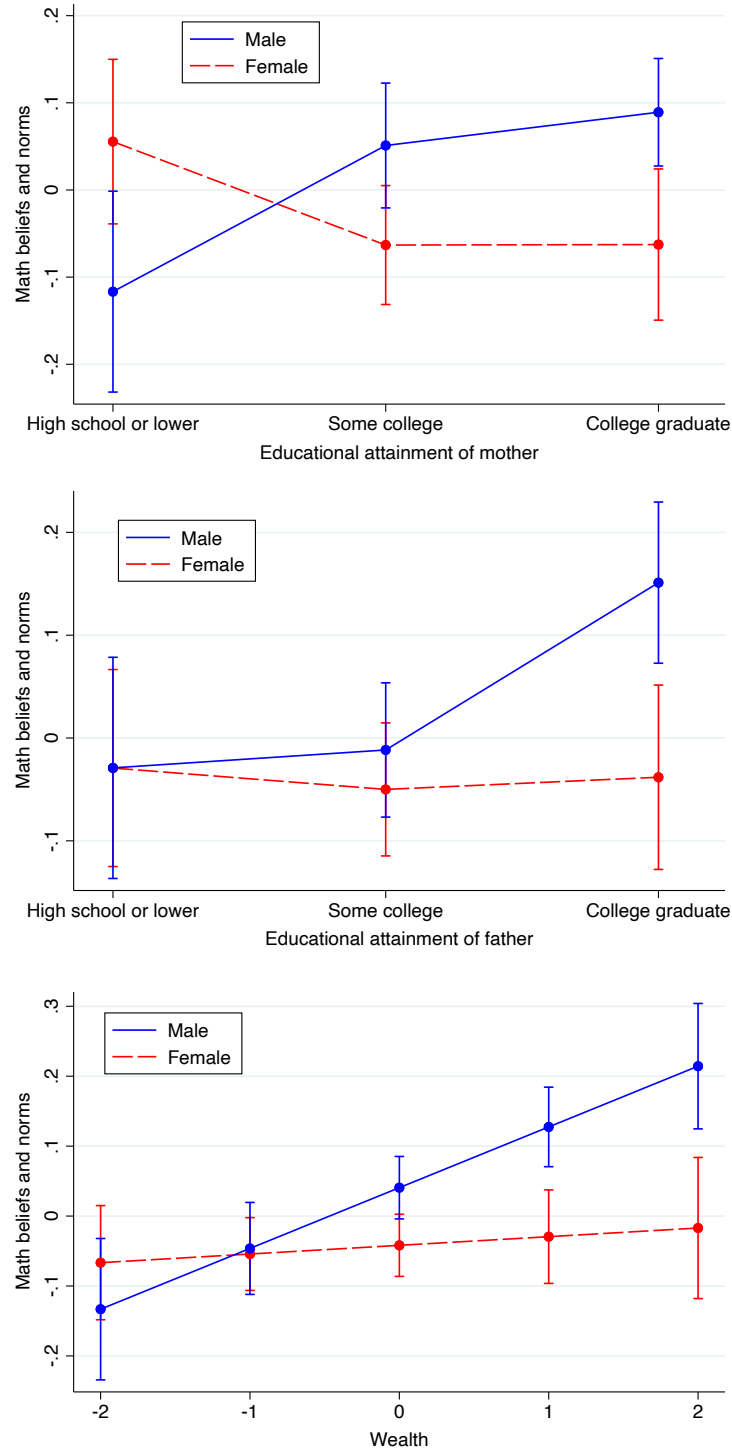


Figure C1. Math Beliefs and Norms by Gender, Parents' Education and Wealth.

The figures show the estimated math beliefs and norms of 15-year old students by gender and across the education level of their mother (top figure), father (middle figure) and wealth (bottom figure), with 90% confidence intervals, based on regression model presented in equation (C1), and estimates presented in Appendix Table C1. The estimates control for the students' math ability.

TABLES FOR APPENDIX C

Table C1. Math Beliefs and Norms, Gender, Parental Education and Wealth

	Math beliefs and norms		
	(1)	(2)	(3)
Female	0.172* (0.095)	-0.0001 (0.081)	-0.082** (0.034)
Some college (Mother)	0.168* (0.092)		
College graduate (Mother)	0.206** (0.086)		
Female $\times$ Some college (Mother)	-0.286** (0.117)		
Female $\times$ College graduate (Mother)	-0.324*** (0.121)		
Some college (Father)		0.017 (0.079)	
College graduate (Father)		0.180** (0.083)	
Female $\times$ Some college (Father)		-0.038 (0.088)	
Female $\times$ College graduate (Father)		-0.189 (0.119)	
Wealth			0.087*** (0.026)
Female $\times$ Wealth			-0.074** (0.036)
Controls	Yes	Yes	Yes
Observations	3,146	2,980	3,206
R-squared	0.025	0.026	0.026

This table shows the estimation results for equation (C1). All models control for students' measured math ability. PISA provides five plausible values of math ability, drawn from a posterior distribution. We standardize these values to have a mean of zero and a standard deviation of one, accounting for the reported student weights from the U.S. survey. Each model is estimated five times, each using a different standardized plausible value as a control, and the final coefficient is the average of these estimates. Standard errors (in parentheses) are computed using a replication method that accounts for PISA's sample design, incorporating both ability measurement uncertainty and standard sampling error (using 80 alternative sets of weights). The Stata command "repest" automates these calculations. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

APPENDIX D

D1 National Survey of Mortgage Originations (NSMO) Survey Questions

We utilize responses from NSMO questions to construct measures of financial literacy and assess the thoroughness of the mortgage search conducted by borrowers. Our analysis focuses on single borrowers (where `borrower_num=1`) who are also the survey respondents (`borrower_r=1`), from waves 1–46 of the NSMO, covering mortgages originated from January 2013 to December 2024.

D1.1 Financial Literacy

We assess the financial literacy of survey participants based on their responses to the survey block X56. Respondents chose their answers from a three-point scale: “Not at all” (coded as 1) to “Very” (coded as 3). The text and codes of the questions, along with the distribution of the responses, are as follows:

How well could you explain to someone the...						
	Code	Response (%)			Score	
		Not	Somewhat	Very	mean	std
Process of taking out a mortgage	x56a	3.8	48.6	47.5	2.41	0.57
Difference btw. a fixed- and an adjustable-rate mortgage	x56b	5.8	27.0	67.2	2.58	0.62
Consequences of not making required mortgage payments	x56f	6.6	26.7	66.8	2.58	0.62
Difference between a prime and subprime loan	x56c	42.8	35.4	21.7	1.74	0.77
Difference between a mortgage’s interest rate and its APR	x56d	25.6	46.2	28.3	1.98	0.73
Amortization of a loan	x56e	27.3	34.1	38.7	2.05	0.81
Relationship between discount points and interest rate	x56h	34.2	37.2	28.6	1.89	0.79

We construct the Basic Literacy measure as the sum of scores from the first set of three questions (x56a, x56b, x56f). Advanced Literacy V1 is the sum of the first three questions in the second set (x56c, x56d, x56e), while Advanced Literacy V2 includes all questions in the second set (x56c, x56d, x56e, x56h). Advanced Literacy V2 is only available for waves 11–30, as question x56h was not included in waves 1–10.

In this block, there are two additional questions—one asking the difference between the lender’s and owner’s title insurance and the other asking the reason payments into an escrow account can change. Since these questions pertain to technical details rather than addressing important decisions and trade-offs, we excluded them from the analysis.

D1.2 Mortgage Shopping Behavior

We evaluate mortgage shopping behavior in four stages: initial familiarity, lender search, loan options, and due diligence. The results are in Table 11.

We assess initial familiarity with the mortgage market based on responses to survey block X05, where respondents choose their answers from a three-point scale: “Not at all” (coded as 1) to “Very” (coded as 3):

When you began the process of getting this mortgage, how familiar were you (and any co-signers) with each of the following?	
	Code
The mortgage interest rates available at that time	x05a
The different types of mortgages available	x05b
The mortgage process	x05c
The down payment needed to qualify for a mortgage	x05d
The income needed to qualify for a mortgage	x05e
Your credit history or credit score	x05f
The money needed at closing	x05g

We construct the Initial Familiarity measure by summing the scores from all the questions in the survey block.

To understand the extent of the lender search, we consider the responses to question x11:

How many different mortgage lenders/brokers did you seriously consider
before choosing where to apply for this mortgage?

We created a Lender Search indicator variable that equals 1 if the borrowers considered more than one lender, and 0 if they did not.

In order to assess whether the borrower is presented with different loan features and options, we used the survey block X23, which has the following questions:

During the application process were you told about mortgages with any of the following?

	Code
An interest rate that is fixed for the life of the loan	x23a
An interest rate that could change over the life of the loan	x23b
A term of less than 30 years	x23c
A higher interest rate in return for lower closing costs	x23d
A lower interest rate in return for paying higher closing costs (discount points)	x23e
Interest-only monthly payments	x23f
An escrow account for taxes and/or homeowner insurance	x23g
A prepayment penalty (fee if the mortgage is paid off early)	x23h

We created a Loan Search variable by adding the number of “yes” responses to these questions.

Finally, we created a Confirm Terms indicator variable if the respondent answered “yes” to question x20h:

In the process of getting this mortgage from your mortgage lender/broker, did you...
Check other sources to confirm that the terms of this mortgage were reasonable.

For the additional analysis in Table D5 we code the responses separately for each question from the X05 survey block. For example, to measure the familiarity with interest rates, we utilize the question x05a:

When you began the process of getting this mortgage, how familiar were you (and any co-signers) with each of the following?...
The mortgage interest rates available at that time.

We create a dummy variable that takes the value of 1 if the answer is “Very” and zero otherwise. We do the same for questions x05b to x05g.

FIGURES FOR APPENDIX D

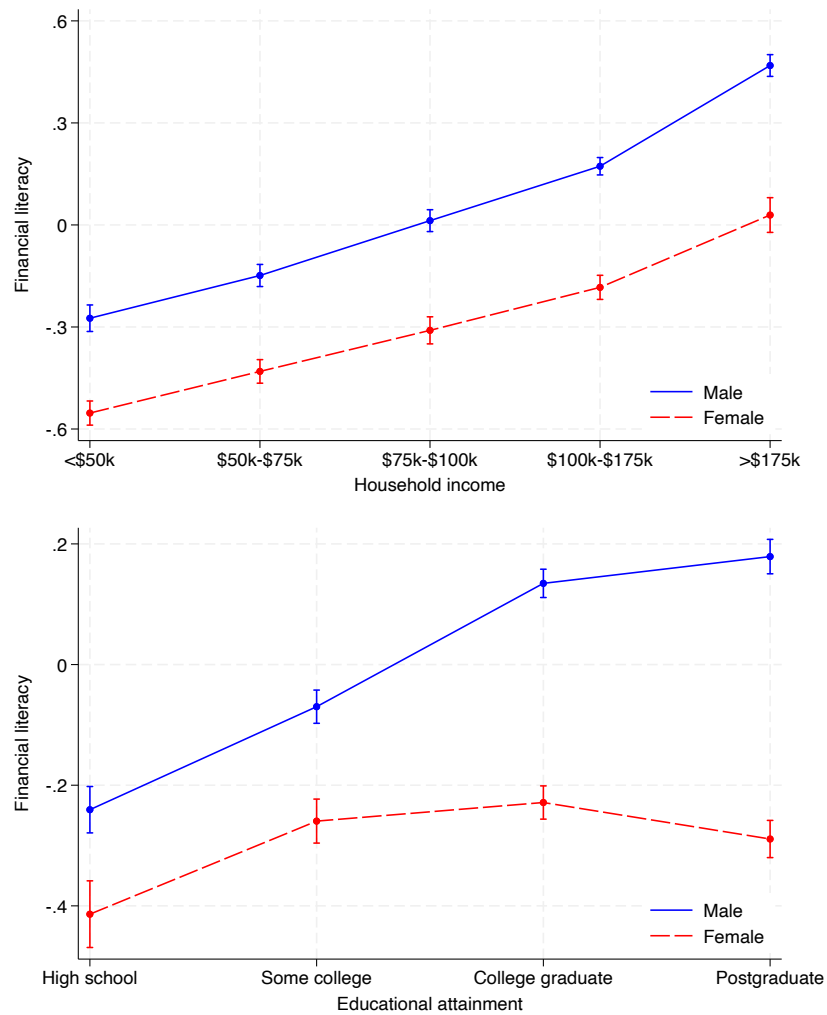


Figure D1. Financial Literacy by Gender, Income, and Education. The top figure displays the estimated financial literacy (Advanced Literacy V1) levels for male and female single mortgage borrowers across various income levels, with 95% confidence intervals. The bottom figure illustrates the estimated financial literacy levels across different educational attainment levels. These estimates are derived from regression models presented in Table 10, by replacing the SES dummy with dummies for five income groups or four education levels.

TABLES FOR APPENDIX D

Table D1. Variable Definitions: NSMO

Variable	Definition	Source
Female	Indicator equal to one if the borrower is female.	NSMO
Income	Total annual household income, reported in categories.	NSMO
Education	Highest educational attainment of the borrower.	NSMO
SES	Indicator equal to one if the household has above-median income and the borrower has completed college education.	Constructed from NSMO
Basic literacy	Average response to the three questions measuring understanding of the mortgage process, fixed- versus adjustable-rate mortgages, and consequences of missed mortgage payments.	Constructed from NSMO
Advanced literacy	Average response to the four questions measuring understanding of prime versus subprime mortgages, APR, loan amortization, and the relationship between discount points and interest rates.	Constructed from NSMO
Initial familiarity	Self-reported familiarity with mortgage interest rates, types and process, credit history, down payment, income and closing fees requirements prior to obtaining the mortgage. Higher values indicate greater familiarity.	Constructed from NSMO
Lender search	Indicator equal to one if the borrower seriously considered more than one lender.	NSMO
Loan search	Indicator equal to one if the borrower seriously considered more than one mortgage type or feature.	NSMO
Confirm terms	Indicator equal to one if the borrower verified the mortgage terms using outside information sources.	NSMO

This table summarizes the variables used in the analyses based on the National Survey of Mortgage Originations (NSMO). Mortgage literacy measures are constructed from borrowers' self-assessments of their ability to explain key mortgage concepts, while shopping variables capture borrowers' search behavior and information acquisition during the mortgage process.

Table D2. Mortgage Literacy, Borrower Gender, and SES: **Single Borrowers with No Reported Spouse or Partner**

	Knowledge Topic									
	Subprime	APR	Amortization	Points	Advanced V1	Advanced V2	Process	Fixed rate	Payment	Basic
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	-0.291*** (0.021)	-0.239*** (0.022)	-0.182*** (0.023)	-0.193*** (0.026)	-0.278*** (0.022)	-0.251*** (0.026)	-0.058** (0.024)	-0.189*** (0.025)	0.018 (0.025)	-0.094*** (0.025)
SES	0.018 (0.059)	0.044 (0.060)	0.041 (0.056)	0.054 (0.066)	0.040 (0.058)	0.046 (0.066)	0.007 (0.058)	-0.095 (0.058)	-0.054 (0.056)	-0.060 (0.059)
Female × SES	-0.165*** (0.047)	-0.112** (0.047)	-0.166*** (0.044)	-0.160*** (0.053)	-0.174*** (0.046)	-0.200*** (0.052)	-0.112** (0.046)	-0.010 (0.046)	-0.036 (0.048)	-0.064 (0.047)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,625	14,625	14,625	10,350	14,625	10,350	14,625	14,625	14,625	14,625
R-squared	0.109	0.075	0.112	0.108	0.123	0.130	0.063	0.088	0.053	0.081

This table re-estimates the specifications from Table 10 using a subsample of single borrowers with no reported spouse or partner who shares household decision-making or financial responsibilities. Robust standard errors are in parentheses. **p<0.05; ***p<0.01.

Table D3. Borrower Sophistication, Borrower Gender, and SES

Borrower sophistication index	
Female	-0.118*** (0.017)
SES	0.042 (0.034)
Female $\times$ SES	-0.082*** (0.031)
Controls	Yes
Observations	29,616
R-squared	0.050

This table reports estimates corresponding to Table 10, using the borrower sophistication index from Bhutta et al. (2026) as the outcome variable. Because the public NSMO data do not include all of the variables used by Bhutta et al. (2026), we construct the index using the subset of their variables that is publicly available. Robust standard errors are in parentheses. *** $p < 0.01$.

Table D4. Mortgage Shopping Behavior, Gender and Financial Literacy:
Single Borrowers with No Reported Spouse or Partner

	Initial Familiarity			Lender Search		
	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.119*** (0.021)		0.021 (0.020)	-0.046*** (0.010)		-0.036*** (0.010)
Financial literacy		0.439*** (0.009)	0.440*** (0.009)		0.034*** (0.005)	0.031*** (0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,625	14,625	14,625	14,625	14,625	14,625
R-squared	0.138	0.280	0.280	0.040	0.042	0.043
	Loan Search			Confirm Terms		
	(7)	(8)	(9)	(10)	(11)	(12)
Female	-0.117*** (0.021)		-0.074*** (0.021)	-0.036*** (0.010)		-0.022** (0.010)
Financial literacy		0.141*** (0.010)	0.135*** (0.011)		0.048*** (0.005)	0.047*** (0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,625	14,625	14,625	11,947	11,947	11,947
R-squared	0.053	0.067	0.068	0.043	0.051	0.051

This table re-estimates the specifications from Table 11 using a subsample of single borrowers with no spouse or partner who shares household decision-making or financial responsibilities. Robust standard errors are in parentheses. **p<0.05; ***p<0.01.

Table D5. Familiarity with Mortgage Concepts, Gender and SES

	Familiarity with specific concepts at the beginning of the search						
	Interest	Mortgage	Mortgage	Down	Income	Credit	Money at
	rates	types	process	payment	needed	history	closing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.067*** (0.008)	-0.058*** (0.008)	-0.016** (0.008)	-0.048*** (0.008)	-0.038*** (0.008)	0.011 (0.008)	-0.032*** (0.008)
Female × SES	-0.031** (0.014)	-0.058*** (0.014)	-0.056*** (0.014)	-0.036** (0.014)	-0.017 (0.014)	-0.013 (0.012)	-0.026* (0.015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	29,616	29,616	29,616	29,616	29,616	29,616	29,616
R-squared	0.112	0.084	0.106	0.089	0.071	0.064	0.069

This table presents the coefficient estimates obtained from regressions of self-reported familiarity with mortgage concepts on a female indicator, an indicator for high SES and their interactions. Robust standard errors are in parentheses. Regressions include survey wave fixed effects, origination year-month fixed effects, dummy variable controls for five income groups, four education levels, five race and ethnicity categories, six age groups and nine credit score categories, and use survey weights. Source: National Survey of Mortgage Originations for conventional loans with a single borrower originated between January 2013 and December 2024. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.